

Behavior Safety of Autonomous Interactive Agents: Risks, Attacks, Defenses and Evaluation

A structured survey of behavioral safety in autonomous interactive agents, spanning intrinsic risks, adversarial attacks, safeguard mechanisms, runtime governance, and evaluation paradigms.

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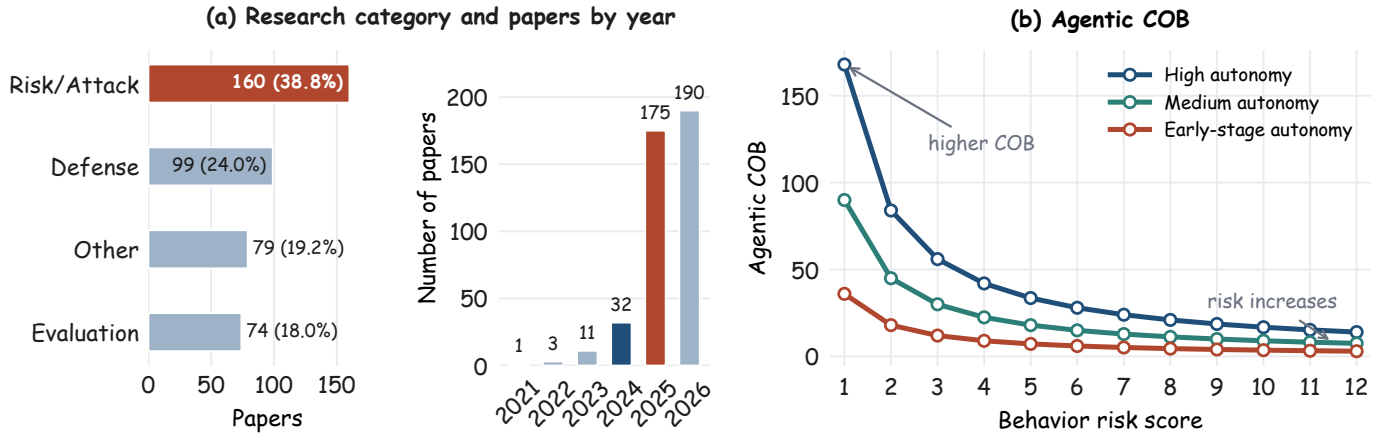
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ABSTRACT

Autonomous interactive agents are capable of long-horizon reasoning, autonomous decision-making, and continuous interaction with open environments and capability evolving. However, existing surveys remain fragmented across individual agent paradigms and safety topics, without a comprehensive understanding of behavioral safety in autonomous interactive agents. In this survey, we present a behavioral safety-centric overview of autonomous interactive agents, covering risks, attacks, defenses, and evaluation. We first investigate the intrinsic behavioral risks arising from insufficient capabilities and design deficiencies in foundation-model reasoning, memory mechanisms, tool use, and multi-agent collaboration, which can gradually deviate from user intentions throughout long-horizon interactions. Building upon these intrinsic vulnerabilities, we further review how adversaries manipulate agent decision-making through data supply chains, user inputs, interaction environments, and collaborating agents, ultimately steering agents toward unsafe or unauthorized behaviors. To systematically understand behavioral safety, we theoretically characterize the coupling between behavioral safety, agent autonomy, and interactivity through the proposed Agentic COB metric. Beyond risks and attacks, we examine how behavioral safety from community ecosystems, including protocols, third-party extensions, and skills, as well as runtime systems, where failures in resource management, isolation, and execution control may undermine system-level safeguards. We correspondingly summarize representative behavioral safety constraints and guidance mechanisms across different threat models, and categorize existing benchmarks from the perspectives of intrinsic behavioral safety and extrinsic behavioral attacks, highlighting the importance of behavioral awareness and trajectory-level evaluation. By establishing a unified behavioral safety-centric perspective, this survey lays the foundation for developing reliable, controllable, and trustworthy autonomous interactive agents and collaborative agent ecosystems.

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ACM XXXX-XXXX/2026/7-ART

<https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

benchmarks from the perspectives of intrinsic behavioral safety and extrinsic behavioral attacks, highlighting the importance of behavioral awareness and trajectory-level evaluation. By establishing a unified behavioral safety-centric perspective, this survey lays the foundation for developing reliable, controllable, and trustworthy autonomous interactive agents and collaborative agent ecosystems.

CCS Concepts: • **Security and privacy** → **Software and application security**; • **Computing methodologies** → **Cognitive science**; **Natural language generation**.

Additional Key Words and Phrases: Autonomous Interactive Agents, Behavior Safety, Inherent Cognitive Vulnerability, Extrinsic Decision Manipulation, Defenses, Benchmark Evaluation

ACM Reference Format:

Xinjie Lai, Weihao Zuo, Zheng Wu, Tianjie Ju, Haodong Zhao, Xiaofeng Zhang, Zhuosheng Zhang, Gongshen Liu, Xinpeng Zhang, and Pengzhou Cheng. 2026. Behavior Safety of Autonomous Interactive Agents: Risks, Attacks, Defenses and Evaluation. 1, 1 (July 2026), 73 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Introduction

The development of artificial intelligence has progressed from static dialog systems to agentic systems capable of complex interaction [1, 195]. As shown in Fig. 1, agents have evolved from early tool-using systems [4, 71, 128, 206, 221, 357] to autonomous workflow agents [98, 230, 250, 321, 408] and further toward memory-augmented [127, 218, 222, 260, 289, 374, 393], interactive [40, 111, 152, 184, 273, 300, 382, 392, 400], and ecosystem-level agent systems. Along this trajectory, agents have continually improved in autonomy, adaptability, and task-solving ability. They employ large language models (LLMs) across diverse modalities as cognitive engines, acquire external knowledge and real-time information through tool invocation, autonomously orchestrate multi-step workflows, and incorporate memory modules to retain historical experience and adapt to user preferences. These capabilities enable agents to interact dynamically with open environments and perform increasingly complex user-instructed tasks. Furthermore, the autonomous interactive agent represented by OS-Atlas [328], UI-TARS [292], Open-AutoGLM¹, Mobile-Agent [294, 360], MAI-UI [400], and WebAgent-R1 [318] have begun to move beyond isolated task execution and toward real-world deployment. By simulating human visual perception, interface understanding, and interactive behaviors, these systems play an increasingly important role in web navigation, code development, mobile automation, and general digital operations [300]. Recently, the emergence of OpenClaw² and Hermes³ has sparked a surge of research on agent runtimes [108, 159, 185]. Autonomous interactive agents are increasingly operating as long-running processes and gradually forming an ecosystem equipped with persistent workspaces, skills, verification loops, and governance. This evolution indicates a broader transition from task-oriented assistants to persistent, personalized, and ecosystem-level operating systems, where agents can continuously learn, collaborate autonomously, and provide proactive support across long-horizon human activities [354].

However, the growing autonomy, coupling environment, and expanding behavioral impact of agents bring new threats whose security implications extend far beyond those of LLMs. Concretely, the compounding complexity of agent architectures introduces intrinsic vulnerabilities—such as tool misuse [253], runtime failures [407], memory conflicts [271], and execution hallucinations [167]—that directly compromise the reliability and trustworthiness of autonomous agent behaviors. These internal fragilities can serve as starting points for designing attack patterns and can amplify harmful results in digital or physical environments, such as initiating malware downloads [389], performing unauthorized asset transfers [85], leaking sensitive information, and making misdirected autonomous driving decisions [181]. Malicious attackers deliberately craft various attack vectors to manipulate agents to

¹<https://github.com/zai-org/Open-AutoGLM>

²<https://openclaw.ai/>

³<https://hermes-agent.nousresearch.com/>

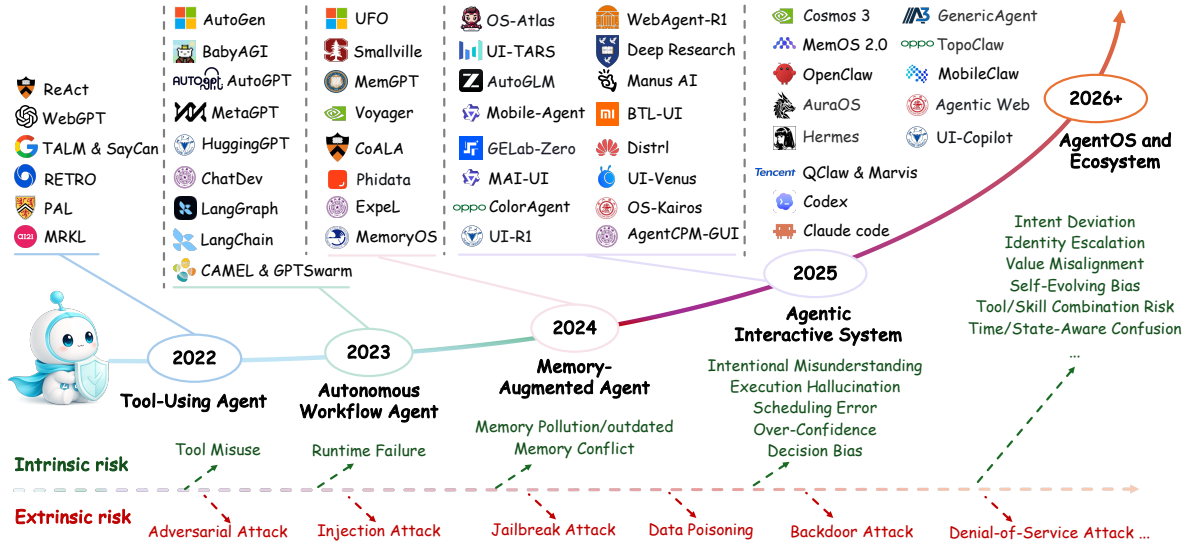


Fig. 1. Evolution of autonomous interactive agents and the expanding landscape of behavioral safety risks. As agents progress from tool use and autonomous workflows to memory augmentation, interactive systems, and emerging AgentOS with its ecosystems, their risk surface expands from localized tool misuse and runtime failures to intrinsic risks rooted in model-system vulnerabilities, and lifecycle-spanning extrinsic attacks.

perform unexpected harmful actions, including adversarial attacks, prompt injection, jailbreak, data poisoning, backdoor attacks, and denial-of-service attacks. Moreover, recent studies highlight severe threats posed by supply chains such as MCP preference manipulation [285], multi-step tool-chain attacks [151], and skill injection [246]. These malicious attacks typically exploit long-horizon interactions, memory persistence, tool invocation chains, reusable skills, and multi-agent coordination mechanisms to compromise the security of an agent’s community execution components [69]. More critically, further exploitation of the aforementioned attack design mechanisms can cause system-level stealthy token exhaustion, privilege escalation, and infinite thinking loop, thereby endangering the entire runtime ecosystem [114]. Consequently, it is imperative to investigate the behavioral safety of autonomous interactive agents from the perspective of their holistic offline and runtime stage compromising internal cognitive vulnerabilities, external loss-of-control risks, community and system safety, which is crucial for the trustworthy deployment and defensive governance of agents.

As a new form of AI products and services, autonomous interactive agents should operate with clearly defined decision-making authority and behavioral boundaries. Mechanisms for behavior tracing and risk warning should be established, and efforts must focus on enhancing the intrinsic security capabilities of agents, thereby mitigating derivative risks arising from their deployment. Although extensive research has explored diverse attack vectors and defense mechanisms, the literature on agent security remains fragmented and exhibits the following limitations.

- The survey of security and safety threats still adheres to a model-centric security paradigm to categorize. For instance, Deng et al. [48] viewed memory as an external resource in light of retrieval, while Yu et al. [366] view memory and tools as internal modules to discuss trustworthiness. Su et al. [259], Ma et al. [192], Wang et al. [295], and He et al. [92] classified and discussed according to system components,

Table 1. Comparison of our survey with existing surveys on autonomous interactive agent safety.

Survey	Taxonomy	Focus	Safety Scope [†]	Def/Eval [‡]	Stage [•]	MAS [◊]	Autonomous
Deng et al. [48]	modular	threat & security	★★★★☆	●○	●●	✓	✓
Yu et al. [366]	modular	trustworthy	★★☆☆☆	●●	●●	✓	✓
Su et al. [259]	capability	autonomy risk	★★☆☆☆	●○	●●	✓	✓
Wang et al. [295]	LLM lifecycle	full-stack safety	★★☆☆☆	●●	●●	✓	✓
He et al. [92]	threat	security & privacy	★★☆☆☆	●○	●●	✓	✓
Lumer et al. [186]	production	tool selection	☆☆☆☆☆	○●	●●	✓	✓
Zhang et al. [391]	design mechanism	memory	★☆☆☆☆	○●	○●	✓	✓
Xu et al. [334]	architecture	skills	☆☆☆☆★	○●	●●	✗	✓
Zhou et al. [399]	infrastructure	harness engineering	☆☆★★★	○●	●●	✗	✓
De et al. [46]	threat	multi-agent systems	★★☆☆☆	●●	●●	✓	✓
Our Survey	safety lifecycle	behavior safety	★★★★★	●●	●●	✓	✓

[†] The four positions denote intrinsic, extrinsic, system-level, and community-level safety, respectively.

[‡] defense (def), and evaluation (eval).

[•] Stage: the two positions denote offline and runtime stages, respectively.

[◊] Multi-agent systems (MAS).

model categories, lifecycle stages, and sources of threats, respectively. Despite extensive research, there remains a lack of attention to a unified safety framework grounded in the behavioral chain, which is vital for the safe execution of increasingly autonomous agents.

- Existing surveys on agent infrastructure tend to focus on functional optimization and threat classification in a fragmented manner. Tool-using surveys concentrated on tool selection [186], memory-augmented surveys concentrated on design mechanisms of memory systems, attack taxonomies, or evaluation benchmarks [56, 105, 323, 391], and skill surveys concentrated on architectural representation and practical deployment [326, 334]. They focused on isolated agent modules and failed to provide a holistic and comprehensive perspective. Although recent works analyzed the interaction mechanisms among memory, skills, and protocols as three coupled forms of externalization [399], and even proposed a completeness matrix consisting of execution environment, tool integration, context management, scope negotiation, loop management, and verification [198], neither of these incorporated the lens from internal safety to external safety.
- Surveys on multi-agent systems primarily center on collaboration mechanisms. For example, Kushwaha et al. [137] and Standen et al. [258] emphasized security issues within the reinforcement learning framework, while De et al. [46], Nguyen et al. [213], Wang et al. [305], and Rafe et al. [236] focused on systemic risks arising from interactions among multiple agents, with classification criteria depending on emergent risks and orchestration topologies.

To bridge the above gaps, our survey systematically delineates a unified behavior-centric safety framework for autonomous interactive agents that spans the entire cognition–execution–environment–collaboration process. Starting from the intrinsic cognitive vulnerabilities of agents, we provide a comprehensive insight into the potential loopholes caused by architectural design or capability deficiencies in single agent execution and multi-agent collaboration, which covertly misalign or deviate behavioral objectives as multi-turn interaction actively progresses. These inherent vulnerabilities further become extrinsic attack surfaces utilized to manipulate deliberate decision-making, including various attack sources, community components, and system-level

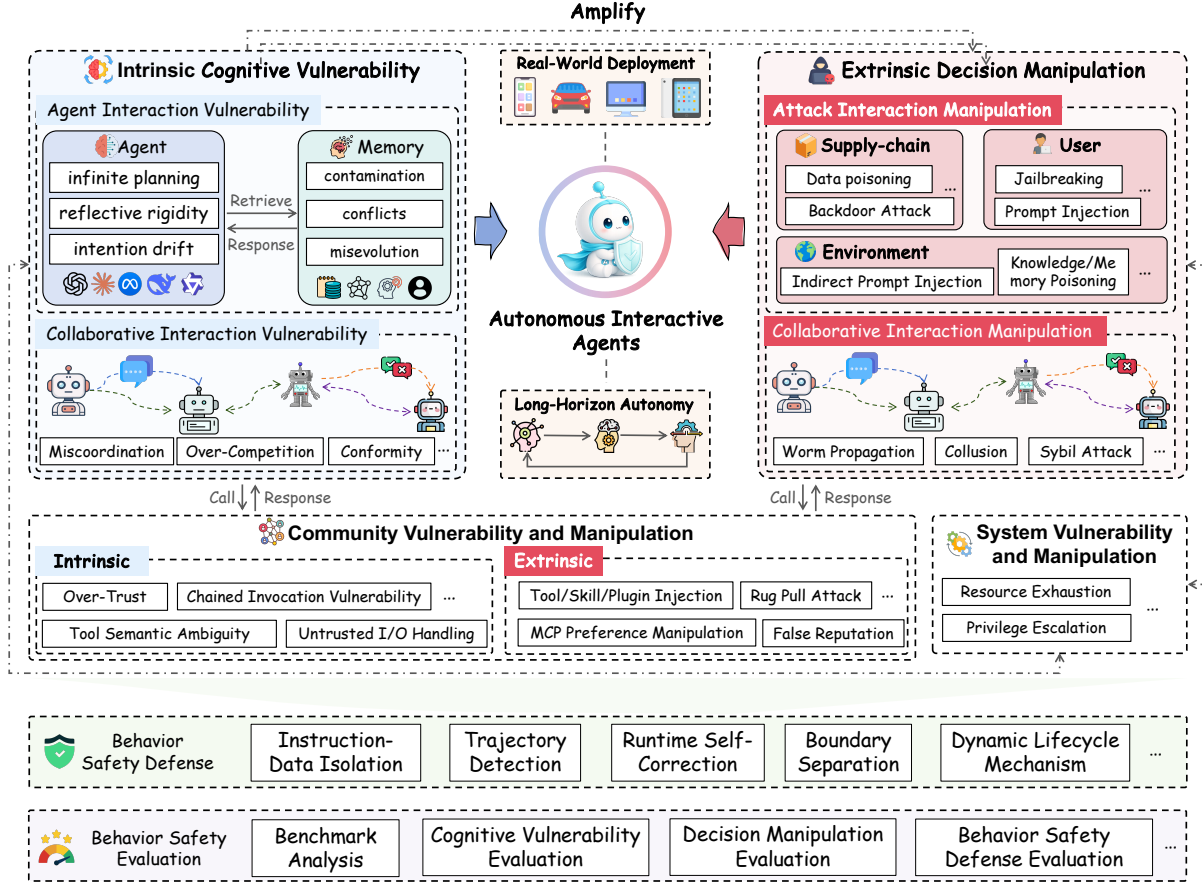


Fig. 2. Overview of our behavior safety taxonomy, featuring risks, attacks, defenses, and evaluation.

governance. Specifically, we recognize and consolidate four external attack vectors: the data supply chain, user input, environment interaction, and cooperating agents, which directly or indirectly mislead agents toward inappropriate or unauthorized long-horizon planning. In light of the increasing operation within open ecosystems, characterizing where behavioral risks originate and how they are exploited is far from sufficient to support the deployment of trustworthy autonomous interactive agents. Thus, we further discuss the safety of interaction components operating in community, encompassing communication protocols, third-party extensions, and skills. Given the reliance of operations on computing infrastructure, we broaden the safety scope to encompass the underlying runtime environment, where failures in resource management, privilege, and system control can undermine safety mechanisms. In addition, under each subcategory, we summarize the associated behavioral safety strategies for the convenience of readers. Finally, we classify existing agent evaluation benchmarks into intrinsic safety and extrinsic attack corresponding to our risk category. By synthesizing the latest literature on agent safety, we offers a comprehensive review and systematic analysis of behavior-centric safety in autonomous interactive agents, as illustrated in Figure 2, expecting to provide a concise roadmap for researchers embarking on agent security studies.

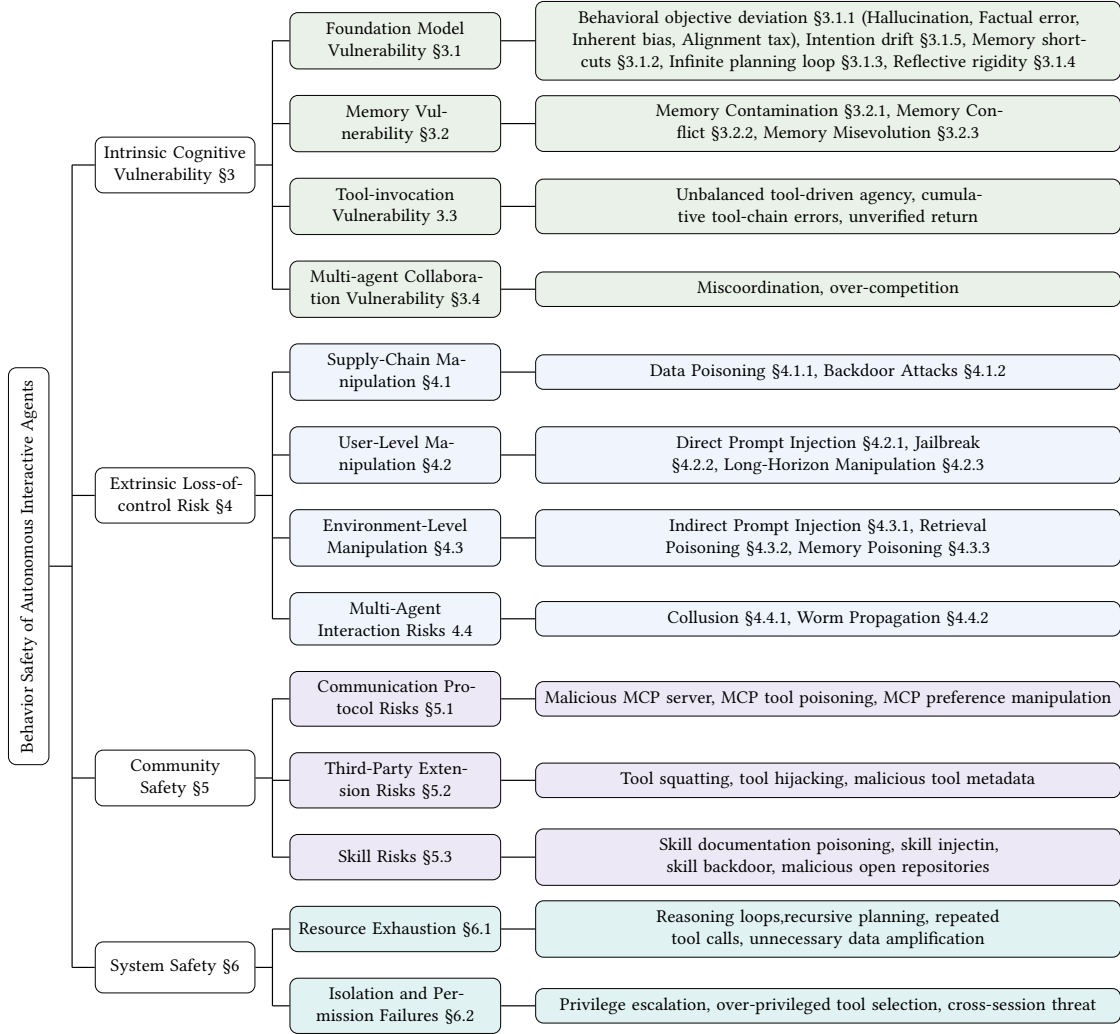


Fig. 3. A taxonomy of behavior safety of autonomous interactive agents.

The main contributions of this survey are summarized as follows:

- We propose a holistic behavior safety framework for the most recent paradigm of autonomous interactive agents, from intrinsic vulnerabilities stemming from architectural design to amplified external security risks. To better recognize the significance of behavior safety, we further introduce a theoretical analysis that measures the trade-off between the capability (autonomy and interactivity) and behavioral risks throughout the entire survey.
- We categorize the endogenous cognitive vulnerabilities of autonomous interactive agents from the lens of foundation models, memory, tool invocation, and multi-agent collaboration, as these internal vulnerabilities arise from flaws in the design mechanisms or capability deficiencies of different modules. Regarding external loss-of-control risks, we summarize malicious manipulations during training and runtime from various

attack sources, including the supply chain, the user, the environment, and other agents. Moreover, we analyze the safety of interaction and execution components in the community of autonomous interactive agents, and further investigate the resource exhaustion, permission failures, and runtime vulnerabilities of agent ecosystem. In addition, we review the latest potential behavioral safety mitigation measures following the aforementioned risks.

- We survey and summarize the existing evaluation benchmarks of behavior safety for autonomous interactive agents and find that even advanced agents still face endogenous cognitive vulnerabilities and external manipulation risks. Additionally, many existing evaluation benchmarks are not comprehensive enough to perform systematic behavior safety analysis for emerging new agent technologies.

This survey emphasizes the radical shift from content safety to behavior safety and seeks to provide a thorough understanding of the behavioral challenges posed by autonomous interactive agents for the development of safer and more trustful self-evolving systems. The remainder of this article is structured as follows. Section 2 introduces the formalization and behavior safety trade-off of autonomous interactive agents, as well as our methodology and survey scope. Section 3 presents the endogenous cognitive vulnerabilities. Section 4 summarizes external loss-of-control risks. Section 5 and Section 6 analyze community safety and system safety. Section 7 synthesizes various existing benchmarks. Section 8 offers future directions for the development of this field. The conclusion of this paper are presented in Section 9.

2 Background

2.1 Autonomous Interactive Agents

We formalize an autonomous interactive agent as a persistent system that proactively interacts with open environments to accomplish long-horizon tasks through autonomous planning, evolving memory, and external tool execution. An agent $\mathcal{A}$ consists of an LLM backbone $\mathcal{F}$ with a system prompt p , an evolving memory bank $\mathcal{M}$, and a tool set $\mathcal{T} = \{T_1, T_2, \dots, T_N\}$ encompassing native tools, reusable skills, and MCP servers. During execution, the agent maintains an internal state s_i encoding the current subgoal, execution history, and other persistent contexts. Given an initial user instruction u or observation o , the agent initializes a task goal

$$g_0 = \Phi_u(u, o_0); \quad (1)$$

and subsequently performs autonomous sequential decision-making without requiring further user intervention. At step i , it receives an environmental observation o_i and produces an action

$$a_i = \pi_\theta(p, g_i, s_i, o_i, \mathcal{T}), \quad a_i \in \mathcal{A}_{text} \cup \mathcal{A}_{tool}, \quad (2)$$

where $\mathcal{A}_{text}$ denotes direct text generation and $\mathcal{A}_{tool}$ denotes tool invocations that alter the digital or physical environment. The observation

$$o_i = (t_i, m_i, q_i) \quad (3)$$

aggregates structured tool outputs t_i , retrieved memory m_i , and supplementary task-specific information q_i . The retrieved memory is formalized as

$$m_i = \mathcal{R}(g_i, s_i, o_i) \in \mathcal{M}, \quad (4)$$

where $\mathcal{R}$ denotes the memory retrieval mechanism.

Following each interaction, the current subgoal, internal state, and memory evolve according to

$$g_{i+1} = \Psi(g_i, s_i, o_i, a_i), \quad (5)$$

$$(s_{i+1}, \mathcal{M}_{i+1}) = \mathcal{U}(s_i, \mathcal{M}_i, o_i, a_i), \quad (6)$$

where Ψ denotes the autonomous planning mechanism and $\mathcal{U}$ updates the agent's internal state and evolving memory according to the execution progress and environmental feedback.

The resulting long-horizon interaction trajectory is

$$\tau = \langle (s_0, o_0) \xrightarrow{a_0} (s_1, o_1) \xrightarrow{a_1} \dots \xrightarrow{a_{H-1}} (s_H, o_H) \rangle, \quad (7)$$

where each action is generated according to the evolving subgoal and internal state rather than additional user instructions. Executing action a_i continuously changes both the external environment and the agent’s internal state, enabling persistent behavioral evolution over long horizons.

The above formulation naturally extends to a multi-agent system $\mathcal{G} = \{\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_K\}$, where each agent maintains its own internal state, memory, and tool set. At step i , the observation of agent $\mathcal{A}_k$ is extended as

$$o_i^k = (t_i^k, m_i^k, c_i^k, q_i^k), \quad (8)$$

where $c_i^k = \{a_i^j, o_i^j \mid j \in \mathcal{N}(k)\}$ denotes the communication messages received from neighboring agents $\mathcal{N}(k)$, including their observations, intermediate reasoning, and execution outcomes. Each agent independently evolves its internal state and memory while collaboratively interacting through the communication channel, jointly producing a multi-agent interaction trajectory.

2.2 Behavior Safety Trade-off

Although recent autonomous interactive agents have achieved remarkable progress in long-horizon reasoning, tool chain use, and proactive environmental interaction, their behavioral safety has not advanced at the same pace. Recent studies consistently reveal that capability improvements are accompanied by increasingly exposed safety failures during autonomous operation [28]. The challenge becomes even more severe when agents continuously interact with complex environments through long execution trajectories. Rather than producing a single response, autonomous interactive agents repeatedly perceive environmental feedback, actively retrieve memories, invoke external tools, continuously update internal states, and make subsequent decisions. Consequently, behavioral risks accumulate throughout the entire interaction process instead of occurring at isolated inference steps [327].

Motivated by these observations, we introduce the metric of Agentic COB, drawing inspiration from the metric design philosophy of Agentic ROI [174]. Agentic COB serves as a unified behavioral safety indicator that characterizes the trade-off between agent capability and behavioral risk during autonomous interaction. Instead of evaluating isolated model outputs, it jointly measures the autonomy, interaction capability, and accumulated behavioral risks exhibited throughout long-horizon execution. We define Agentic COB as:

$$\begin{aligned} \text{Agentic COB} &= \frac{\text{Autonomy} \times \text{Interactivity}}{\text{Behavior Risk}} \\ &= \frac{\text{Trajectory Length} \times \text{Interaction Turn}}{\text{Risk Score}}. \end{aligned} \quad (9)$$

Specifically, Agentic COB computes the compound value derived from the agent’s ability to accomplish long-horizon tasks without human-in-the-loop while maintaining the robustness and resilience against unexpected disturbance or intentional disruption. A higher COB indicates better autonomy and interactivity of safe and trustworthy agents. The formulation is composed of three components:

- **Autonomy** captures the length of executable trajectories which are related to the successful completion of goal composition, multi-step reasoning, long-horizon planning, deliberate decision-making, execution, and self-reflection. Obviously, the longer the trajectory, the greater the autonomous capability.
- **Interactivity** refers to the secure, effective interaction turns with external environments facilitating further progress toward the final goal. This includes perception, semantic or visual grounding, memory retrieval, tool chain invocation, skill calls, as well as communication and collaboration. Excellent interaction capability is significant for agents to accomplish user-specified tasks in diverse and dynamic environments.

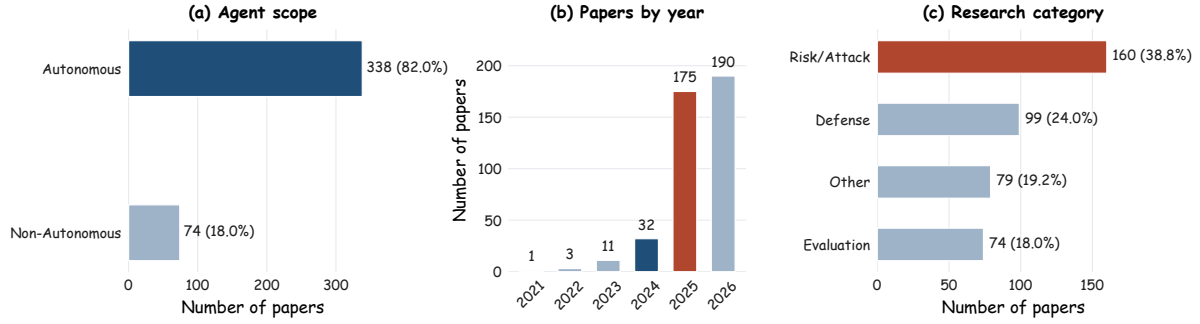


Fig. 4. Statistics and distribution of the literature included in this survey. Left: distribution across agent types. Middle: publication distribution by year. Right: distribution across research categories, including risk/attack, defense, evaluation, and others. The “Others” category primarily comprises survey papers and foundational research on agent architectures.

- **Behavior Risk** is measured by the score in various failures encompassing flawed planning or reasoning, objective deviation, manipulated decision-making, deceptive reflection, memory misevolution, improper tool use, over-privileged or unconfirmed action, delegation or cooperation breakdowns, and insecure or wasteful execution. These risks may further lead to privacy leakage, sensitive data exposure, budget drain, supra-competitive pricing, brittle generalization, and unsafe operations such as downloading malware and transferring digital assets.

Runtime execution is where Agentic COB becomes materially realized. Throughout long interaction trajectories, autonomy and interactivity continuously expand the agent’s operational capability, while behavioral risks simultaneously accumulate through reasoning errors, unsafe tool invocation, memory corruption, objective deviation, or malicious environmental influence. Consequently, improving agent capability without proportionally strengthening runtime safety inevitably enlarges the behavioral attack surface. Therefore, ensuring trustworthy autonomous interaction requires safety mechanisms to evolve alongside increasingly capable agents. Rather than relying solely on input-level filtering, future defenses should provide continuous runtime protection through trustworthy memory management, secure tool invocation, execution monitoring, provenance verification, and cross-lifecycle policy enforcement. Overall, Agentic COB provides a principled perspective for understanding the capability–risk trade-off of autonomous interactive agents and serves as the conceptual foundation for the behavioral safety taxonomy presented in the remainder of this survey.

2.3 Methodology and Scope

This survey focuses on the emerging paradigm of autonomous interactive agents, namely AI systems that continuously interact with open environments through long-horizon reasoning, persistent memory, external tool invocation, reusable skills, and multi-agent collaboration. Accordingly, the surveyed literature encompasses a broad spectrum of agent paradigms, including general LLM agents, web agents, GUI agents, mobile agents, embodied agents, software engineering agents, and recently emerging Agentic Web and ecosystem-level runtime systems. They have evolved from isolated task execution toward continuously operating autonomous ecosystems, where increasingly complex interactions fundamentally reshape the behavioral risk landscape.

Unlike previous surveys that organize the literature according to attack categories, architectural modules, or system lifecycle stages, this survey adopts a behavior-centric perspective to characterize how safety risks impact autonomous execution. Our scope extends beyond conventional adversarial attacks to encompass the entire behavioral safety of autonomous interactive agents. Specifically, we theoretically characterize the coupling

between behavioral safety, agent autonomy, and interactivity through the proposed Agentic COB metric, serving as the conceptual foundation for the behavioral safety taxonomy presented in the remainder of this survey. We first investigate intrinsic cognitive vulnerabilities originating from the capability or design limitations of foundation models, memory mechanisms, tool invocation, and multi-agent collaboration, gradually eroding behavioral objectives and user intentions throughout long-horizon interactions. Building upon these inherent vulnerabilities, we further review the externally induced decision-making manipulation through data supply chain, user inputs, environmental interactions, and collaborating agents, which ultimately steer agents toward unsafe or unauthorized behaviors. As autonomous agents increasingly rely on shared protocols, third-party skills, and persistent runtime infrastructures, we further broaden the discussion to community safety and system safety, covering communication protocols, reusable skills, runtime isolation, privilege management, isolation boundary, and execution governance. Correspondingly, we summarize representative behavioral safety constraints and guidance frameworks, as well as evaluation benchmarks from the perspective of intrinsic behavioral safety and extrinsic behavioral attacks, highlighting the significance of behavioral awareness and trajectory-level evaluation. By delineating a unified behavioral safety framework, we hope to foster the development of reliable, controllable, and trustworthy autonomous interactive agents and collaborative agent ecosystems.

Our methodology is structured as follows. First, we conducted a keyword-based search targeting agent types and risk types to identify relevant papers, mostly from January 2025 to July 2026. Next, we manually filtered out non-behavior-related and non-safety papers. For each remaining paper, we categorized its proposed method or framework by analyzing its risk, attack, defense, evaluation types based on the behavioral safety of intrinsic cognition and extrinsic execution, assigning them to appropriate categories and subcategories. Totally, we reviewed 412 technical papers, with their distribution across years, agent types, and risk/defense/evaluation/other strategies illustrated in Figure 4, highlighting the importance of ensuring that the long-horizon operation of autonomous interactive agents remains under human control at all times. As shown, behavior safety research on autonomous interactive agents has surged remarkably since 2025, especially following the release of OpenClaw. In addition, we primarily collected literature from major venues spanning machine learning, natural language processing, computer vision, software engineering, systems, and cybersecurity, including ACM Digital Library, IEEE Xplore, SpringerLink, USENIX, arXiv, and top conferences such as NeurIPS, ICML, ICLR, ACL, EMNLP, CVPR, AAI, IJCAI, ICCV, ECCV, WWW, and CCS.

3 Intrinsic Cognitive Vulnerability

Integrated advanced LLMs as core internal engines, recent autonomous interactive agents can proactively interact with dynamic environments, iteratively process feedback, and make deliberate real-time decisions [48]. However, as these agents tackle increasingly complex long-horizon tasks, their heightened autonomy and interactivity expose a growing set of inherent cognitive vulnerabilities that directly threaten behavioral safety [138]. Unlike earlier LLM-driven agents that primarily faced static output errors, contemporary agents also suffer from emerging risks rooted in the design of their architectures or the insufficient cognitive capabilities vital for safe extended operation. Drawing on the viewpoint of He et al. [92], agents not only inherit the endogenous cognitive vulnerabilities from foundation models, but also exhibit specific intrinsic safety vulnerabilities introduced by additional components—memory, tool invocation, and multi-agent collaboration—which serve as the categories discussed below and illustrated in Figure 5.

3.1 Foundation Model Vulnerability

In autonomous interactive agents, foundation model vulnerabilities manifest as distinct behavioral safety failure modes during proactive environment perception, task decomposition, long-horizon planning, and deliberate reasoning. Unlike earlier systems where such flaws largely remained at the output level, contemporary agents

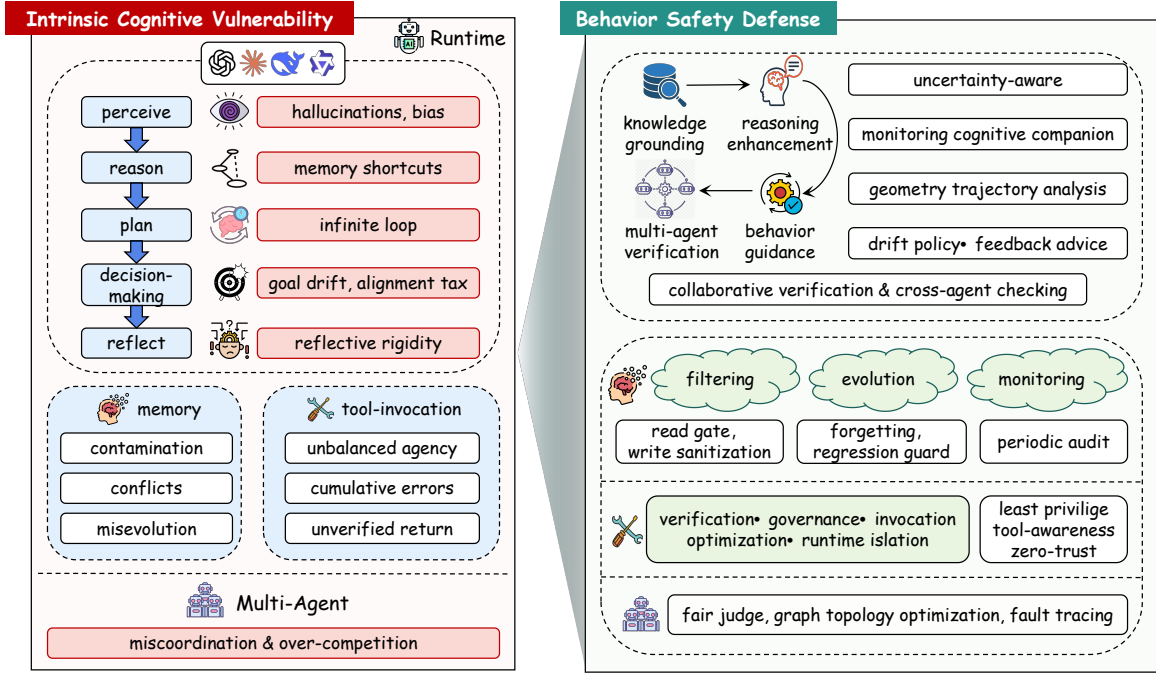


Fig. 5. The endogenous cognitive vulnerability of autonomous interactive agents with corresponding demonstrations on attack and behavioral defense.

exhibit behavioral objective deviation, intention drift, memory shortcuts, infinite planning loops, and reflective rigidity, which all directly erode the safety and reliability of autonomous behavior.

3.1.1 Behavioral Objective Deviation. Behavioral objective deviation arises when the behaviors generated by foundation models fail to satisfy the operational objectives required for safe and robust autonomous interactive agents. These deviations may compromise the truthfulness, reliability, fairness, or utility of agent behaviors, resulting in unsafe or misaligned interactions with users and external environments [204]. Recent research has primarily identified four representative manifestations of behavioral objective deviation: hallucinations, factual errors, inherent bias, and alignment tax.

❶ **Hallucination** refers to fabricated, logically inconsistent, or erroneous outputs, which are often spontaneously generated within the agent’s own cognitive processes [167]. Unlike LLM hallucinations those are primarily confined to linguistic errors, agent hallucinations can emerge throughout the entire agent pipeline and propagate deviations along the long-horizon trajectory [176]. For example, planning hallucinations generate steps that cannot be executed in reality, tool hallucinations fabricate the return value of a non-existent function, and reflection hallucinations continue to frequently repeat the exact same errors [315], despite being capable of generating convincing self-criticism. In embodied or autonomous settings, these hallucinations may further lead to incorrect task execution, unsafe behavioral operations, unreliable interactions in the real world, thus degrading overall reliability.

Behavioral Safety Defense. By leveraging models’ ability to adapt to new tasks without parameter updates, In-Context Learning offers a promising direction for mitigating hallucinations [160]. Vu et al. [283] showed that incorporating up-to-date evidence retrieved from search engines into prompts could improve factual accuracy

and reduce hallucinations. Beyond prompt-based methods, reasoning-oriented approaches mitigate hallucinations by structuring and verifying the reasoning process. Zhao et al. [396] proposed Logical Chain-of-Thought, which guided models to reason through explicit and verifiable logical steps, while Dhuliawala et al. [50] introduced Chain-of-Verification, enabling multi-step self-verification and iterative revision of generated responses. Furthermore, Liang et al. [162] proposed RLKF, which trained models to estimate the uncertainty of their own reasoning by learning the relationship between hidden representations and answer correctness. Building on this idea, Xu et al. [335] encouraged models to explicitly generate self-reflective rationales and confidence estimates, incorporating reliability awareness into the generation process. Recent research has further extended hallucination mitigation to multi-agent collaboration paradigms. Gosmar et al. [77] constructed a multi-agent orchestration mechanism based on the OVON framework. Through process-level constraints and natural language-based content recalibration during multi-turn collaboration, the framework continuously filtered and corrected intermediate errors, thereby reducing the propagation of hallucinations across the reasoning pipeline. Furthermore, Pan et al. [220] developed a multi-agent framework for ophthalmic diagnosis, where a Decision Agent dynamically selected and orchestrated external tools to construct a traceable reasoning workflow. Combined with cyclic tool verification, closed-loop auditing, and iterative error correction, the framework reduced hallucinated diagnoses while improving the reliability and traceability of decision-making. As agent architectures become increasingly complex, Alansari et al. [6] argued that effective hallucination mitigation requires complementary defenses targeting different failure modes, rather than redundant combinations of similar strategies. In particular, integrating reasoning-based correction with external evidence alignment can provide more robust improvements in factuality and faithfulness.

② **Factual error** is the semantic deviation where the agent’s output conflicts with objectively verifiable facts, focusing on the assessment of correctness at the outcome level. The erroneous information may be propagated increasingly when users trust the reliability of generated content, thereby amplifying its negative impact. In medical contexts, this can further lead to the dissemination of misinformation that adversely affects individual clinical decisions and broader public health outcomes [201]. By contrast, from the perspective of generative mechanisms, hallucination tends to characterize the phenomenon where a model produces information when lacking the constraints of reliable evidence. Consequently, we discuss factual error as a distinct risk associated with foundation models that can be analyzed independently.

Behavioral Safety Defense. To automatically detect the factual errors, Cohen et al. [43] proposed a zero-shot cross-examination framework, through prompting-based multi-turn interactions between the examinee and examiner, the method exposed semantic inconsistencies and unsupported statements, hence improving factual error detection. Beyond self-verification mechanisms, recent studies have explored retrieval-grounded safety guidance to mitigate factual errors in agent outputs. Li et al. [145] proposed Retrieval-Augmented Correction, which retrieved external evidence to identify and revise unsupported statements after generation without requiring additional model fine-tuning. Further extending toward memory-aware generation frameworks, Chen et al. [31] introduced Explicit Working Memory, which continuously updated an external memory module using online fact-checking feedback during long-form reasoning and generation. By dynamically revising unsupported claims throughout the reasoning trajectory, the framework reduced factual drift and improved consistency across extended agent interactions.

③ **Inherent bias** in agentic systems refers to systematic and unfair favoritism or prejudice, which can lead to discriminatory and stereotypical outputs. The use of unrepresentative data, the inappropriate use of AI algorithms in model development and the improper user interactions with the model are three factors responsible for inherent biases in agents. Such flaws can be amplified and embedded into automated decision-making pipelines in real-world applications, leading to inconsistent or even discriminatory behavior against specific groups. A recruitment model may screen out qualified candidates due to gender or racial biases, while a medical diagnostic system may yield missed or erroneous diagnoses for minority populations [91].

Behavioral Safety Defense. To reduce biased inputs at the source, Hanna et al. [89] emphasized mitigating bias through joint interventions across the data, model and application layers, specifically by improving the representativeness and annotation fairness of training data. Recent studies have also explored self-debiasing guidance mechanisms to suppress biased outputs. Gallegos et al. [68] proposed a zero-shot self-debiasing technique that enables large language models to recognize and iteratively reduce stereotypical generations through self-reflection prompts, without modifying model parameters or training data. By incorporating internal bias-aware feedback into the decoding process, the framework lightweight behavioral guidance for safer and fairer text generation. Further extended into multimodal agent systems, Cheng et al. [38] proposed a social debiasing guidance framework by introducing counter-stereotype guidance during multimodal alignment and generation to continuously steer the model away from biased semantic associations. Beyond single-model guidance, Owens et al. [217] proposed a multi-LLM debiasing framework in which multiple language models jointly evaluate and refine generated outputs to suppress implicit negative stereotypes. This cooperative feedback mechanism improved fairness robustness and demonstrated the potential of distributed safety guidance for mitigating bias propagation in agent-based systems. Xu et al. [337] proposed a multi-objective approach within a multi-agent framework. MOMA deployed a masking agent to identify and suppress potential bias-related features, as well as a balancing agent to introduce neutral or rephrased vocabulary during generation. This closed-loop optimization over generation and feedback progressively mitigated socially biased outputs, and demonstrated the potential of causal interventions for debiasing.

④ **Alignment tax** refers to the intrinsic trade-off between safety alignment and agent utility, where improving safety capability often comes at the cost of reasoning degradation [110], execution impairment, and over-refusal behaviors in autonomous agents.

Behavioral Safety Defense. The research of behavioral safety defense against alignment tax mainly concentrate on reasoning and trajectory. Deliberative Alignment [81] explicitly retrieved and reasoned over safety specifications prior to response generation, thereby shifting safety alignment from implicit policy imitation toward interpretable deliberative reasoning. Through policy-grounded compliance assessment before action generation, the framework more effectively differentiated harmful and benign requests, thereby reducing over-refusal behaviors while maintaining benign task utility. Later research begun to leverage trajectories to help better balance safety and utility. The SafeConstellations [194] dynamically captured task-specific internal representational trajectories during the inference stage, and precisely steered requests with benign intent that would otherwise trigger safety mechanisms from a refusal pathway toward a normal execution pathway. AgentAlign [380] leveraged abstract behavior chains as cognitive guidance to construct benign agent trajectories while incorporating sensitive edge-case scenarios to calibrate the boundary between harmlessness and helpfulness, thereby preserving benign task executability under safety constraints. Furthermore, Failure-Trajectory Evolution(FATE) framework [364] adopted an online policy optimization perspective, and employed Pareto Front Policy Optimization to select corrective samples from agent-generated failure trajectories that jointly balance safety, utility, and refusal calibration. It transformed multi-objective constraints into effective signals for policy evolution, substantially improving safety while preserving the agent’s task execution performance.

3.1.2 Memory Shortcuts. Memory shortcuts reveal that agentic systems often behave as lazy learners [268], directly reusing superficially relevant past experiences instead of performing semantic reasoning and decision-making. For example, upon detecting the token “Starbucks”, a personal assistant agent may retrieve a prior behavioral pattern such as “purchase an iced Americano” without re-evaluating the user’s current intent or context. This recall relies on associative correlations along historical trajectories rather than causal environmental structures [55], leading to poor generalization under out-of-distribution scenarios. On the one hand, as memory frequency and recency accumulate, agents tend to over-rely on specific interaction history, which reduces their adaptability to newly emerging contexts and dynamic environments [100, 211]. On the other hand, through

iterative feedback and memory updating mechanisms, these spurious correlations can become progressively self-reinforcing [267]. Actions induced by shortcut memories generate new trajectories that further strengthen the same biased associations, ultimately forming a closed feedback loop that amplifies decision bias and undermines robust reasoning of autonomous interactive agents.

Behavioral Safety Defense. To alleviate the shortcut dependency on specific memories, existing defense methods approach the problem from dimensions including distributed episodic memory control, trajectory-based selective retrieval, causality-calibrated memory frameworks, and lightweight runtime self-defense. Larimar [45] designed a distributed episodic memory controller, which maintained contextual consistency and experiential controllability during long-range reasoning. Utilizing agent’s trajectory, HIAGENT [103] introduced a trajectory retrieval module that selectively recalled complete historical action-observation pairs during long-horizon task execution. When sub-goal execution failed, the detailed trajectory information could facilitate failure attribution and corrective replanning, while previously successful trajectories provided structured experiential guidance for decision-making in novel scenarios. Tang et al. [267] designed a plug-and-play causality-informed memory calibration framework, which filtered context-dependent noisy memories during the write phase and re-ranked candidate memories according to causal invariance and cross-environment stability during retrieval. Moreover, Neelou et al. [209] proposed A2AS, a lightweight runtime security layer similar to how HTTPS secured HTTP. By implementing the BASIC security model—which comprised behavior certificates, authenticated prompts, security boundaries, in-context defenses, and codified policies—it activated model self-defense via leveraging the model’s own reasoning capabilities, and all without adding latency overhead or external dependencies.

3.1.3 Infinite Planning Loop. Autonomous interactive agents that iteratively replan based on execution feedback may get stuck in infinite planning loops, repeatedly generating ineffective or redundant action sequences without making substantive progress toward user assignment. This pathological planning state prevents effective corrective refinement and results in excessive step consumption, stalled execution trajectories, and severe degradation in long-horizon task performance [329, 361]. Recent studies suggest that lengthy contextual histories and noisy intermediate feedback can substantially affect the agent’s planning and replanning capabilities, limiting its ability to generate corrective strategies [329]. More importantly, agents have been found to frequently enter repetitive planning cycles or select non-progressive actions even when explicitly instructed to avoid invalid operations, revealing intrinsic limitations of current foundation models in adaptive sequential decision-making [245].

Behavioral Safety Defense. Current safety strategies focus on hard termination mechanisms that abruptly halt execution, sacrificing productive reasoning [130]. For instance, KnowLoop [398] proactively identified unreliable decision steps, aborted execution, and requested human intervention when the uncertainty exceeded the threshold. The LoopGuards strategy [72] provided explicit termination control by setting a maximum number of iterations, a deadline, and a time budget, thereby ensuring the convergence of the planning process. And Schepanowski et al. [245] designed a dedicated loop detection mechanism, which treated an infinite loop as a termination reason. Its specific feedback modular informed the model at which action it became trapped in the loop, and instructed it to restart from the state prior to the loop. However, purely rule-based termination mechanisms often interrupt potentially recoverable reasoning trajectories, limiting agent autonomy and task completion capability. To address this limitation, recent studies have shifted toward planning-aware mitigation strategies that proactively reduce the likelihood of agents entering infinite planning loops. Li et al. [144] proposed AOP for multi-agent systems, in which a reward model ensured that every decomposed subtask was within the agent’s capabilities to prevent it from getting stuck in planning unsolvable subtasks. Khan et al. [130] introduced the Cognitive Companion, a parallel monitoring architecture with an LLM-based Companion. Like a thinking partner who remained silent during productive work but intervened constructively when obstacles arose, the Companion preserved agent autonomy while providing targeted assistance when detected infinite planning loop.

3.1.4 Reflective rigidity. Reflection is a post-hoc correction and self-refine process [247], in which autonomous interactive agents check, evaluate and rectify their reasoning, planning, and actions. The cognitive rigidity of reflection manifests itself as high confidence in previous conclusion when self-evaluating, and adherence to early flawed cognition. For instance, Weatherhead et al. [315] indicated that GPT-o4-mini offered a stubbornly inaccurate feedback choice rather than delivering a targeted correction in the search and identification task. This inherent rigidity hinders agents from adjusting subsequent strategies based on reviewing and summarizing their experiences and also allows errors to persist in the following reflection loops.

Behavioral Safety Defense. Existing defenses against reflection-induced cognitive rigidity primarily improve reflective adaptability by incorporating external behavioral verification, diversified reasoning, adaptive cognitive regulation, and state-aware reasoning intervention. Weatherhead et al. [315] argued that reflective reasoning in contemporary agents should be grounded in executable guardrails, particularly when intrinsic metacognitive capabilities remained insufficient. Instead of relying solely on self-generated reflective narratives, executable guardrails introduced externally verifiable feedback loops that enforced constraint-aware correction and execution-level validation. This paradigm shifted reflection from a purely linguistic process toward one grounded in behavioral repair and iterative reasoning refinement. Unlike executable guardrails that depend on external verification signals, Self-Contrast [247] improved reflection through internally diversified reasoning dynamics. Through comparative evaluation across divergent reasoning trajectories, the framework exposed reasoning discrepancies and facilitated more accurate and stable self-correction. Tsui et al. [281] incorporated subtle cues such as the word "Wait" to significantly enhance an agent's ability and reliability in self-correcting output errors in output. Cognition-aware Policy Optimization [348] further alleviated rigid reflective reasoning by introducing adaptive step-level cognitive-depth allocation. Lee et al. [142] proposed Spike-Triggered Adaptive Reasoning Steering, a training-free framework designed to rectify cognitive rigidity by monitoring latent dynamics. STARS employed geometric trajectory analysis to diagnose whether the agent was stubbornly adhering to a flawed yet internally consistent reasoning trajectory. The method then injected state-aware language cues to steer the agent away from entrenched cognition and back toward correct logic.

3.1.5 Intention Drift. Intention drift refers to the inconsistency with initial human-assigned constraints and instructions as multiple rounds of interaction proceed. This unintended deviation can self-consolidate and self-reinforce through the dynamic feedback loop of autonomous interactive agents, bringing potential drift in behavior. Based on an analysis and summary of the literature reviews, we outline the following intention deviation categories. ❶ **Goal deviation.** Arike et al. [10] found that agents exhibited tendency to gradually deviate from their originally specified objectives during prolonged interactions or under competing goals. Such goal drift became increasingly severe as interaction horizons and contextual complexity grow, revealing the limited capability of current agents to consistently preserve long-term objectives. ❷ **Semantic drift.** Yu et al. [367] observed that tool-enabled agents spontaneously developed multi-step strategies to circumvent safety constraints. Even if the outputs remain syntactically valid, the underlying decision-making logic has deviated from the initial configuration [242], demonstrating the inherent structural challenges in maintaining the consistency of execution-oriented behavior during decision-making process [239]. ❸ **Multi-agent divergence.** Han et al. [88] identified a critical post-deployment risk termed Alignment Tipping Process, in which the continual self-evolution of LLM agents gradually abandoned their initial alignment constraints under environmental feedback. In multi-agent settings, such violations could further diffuse across interacting agents [15], ultimately leading to persistent divergence from human intent and collective misalignment [12]. Besides, role drift occurred when agents gradually deviated from their designated responsibilities and began to perform actions outside their expertise in extended collaboration [288]. For instance, a coder agent might start providing feedback, while a reviewer agent might begin writing implementation code, leading to degraded overall performance.

Behavioral Safety Defense. The consequences of intention drift are particularly severe in long-running autonomous interactive tasks. Without proper mechanisms to detect and correct these deviations, risks compound across interaction turns, ultimately causing behavioral failure or requiring significant human intervention. To mitigate intention drift, it is necessary to reliably detect and quantify the extent of drift. Residual Knowledge Stability [242] introduced a systematic framework to evaluate semantic consistency and residual knowledge preservation across sequential tasks. By quantifying accumulated semantic drift during long-horizon interactions, it provided interpretable signals for intention drift detection and semantic stability analysis. Becker et al. [15] further proposed DRIFTJudge and DRIFTPolicy, where the former identified behavioral drift at test time and the latter provided feedback suggestions, enabling agents to self-calibrate discussions and return to initial objectives. However, detection alone is insufficient for correcting behavioral drift. Rath et al. [239] proposed RoleFix, which combined rule-based and semantic drift detection with a reflection–reassignment–resumption cycle for automatic correction. Su et al. [259] introduced the Reflective Risk-Aware Agent Architecture, integrating risk assessment, behavioral reflection, and policy verification into the decision pipeline to enable continuous safety monitoring and adaptive action adjustment. Beyond task-level consistency, deployed agents must also align with users’ evolving implicit intentions. Liu et al. [190] proposed PersonalAlign, a hierarchical implicit intent alignment framework for personalized GUI agents. By leveraging long-term user-centric records, it guided agent behaviors according to evolving user preferences and routines, enabling proactive assistance. During multi-turn interactions, even minor errors or stylistic deviations of an agent can be continuously amplified through the use of its own outputs as subsequent inputs, thereby forming a negative feedback loop. Adaptive Behavior Anchoring [239] addressed this issue by injecting dynamic baseline examples to redirect agents toward original task specifications, preventing the accumulation of behavioral deviations over long horizons.

Note—Capability–Safety Trade-off of Foundation Model

Foundation model vulnerabilities reveal that behavioral safety in autonomous interactive agents originates from the intrinsic reliability of their cognitive processes rather than isolated output correctness. Hallucinations, intention drift, memory shortcuts, infinite planning loops, and reflective rigidity progressively accumulate across long-horizon interactions, amplifying behavioral deviations through repeated reasoning, planning, execution, and reflection. Recent behavioral safety research has shifted from correcting individual responses toward regulating the entire behavioral trajectory through reasoning guidance, memory calibration, planning intervention, and reflection-aware optimization. From the perspective of Agentic COB, these defenses aim to maximize autonomous capability and interaction effectiveness while suppressing the accumulation of behavioral risk throughout execution. This highlights a fundamental trade-off in agent design: increasing autonomy without simultaneously improving intrinsic behavioral robustness inevitably enlarges the cumulative risk of long-horizon autonomous interaction, making behavioral reliability a prerequisite rather than a by-product of advanced agent capabilities.

3.2 Memory Vulnerability

In autonomous interactive agents, memory systems function not merely as passive stores but as active shapers of behavioral planning and decision-making. As trajectory length and interaction turns scale up, the memory mechanisms that sustain long-horizon autonomy may suffer from memory contamination, memory conflicts, and memory misevolution, which can accumulate silently across continuous interactions, transforming benign recall into a source of behavioral instability.

3.2.1 Memory Contamination. During ordinary interactions of autonomous agents, previously accumulated benign information can silently mislead unrelated tasks and negatively influence user-facing decision-making and

behavior without attack-driven prompt injection, which we refer to as memory contamination. Once seemingly plausible yet unverified memories enter the context window, standard memory consolidation mechanisms may inadvertently promote short-term pollution into enduring long-term memories [170]. As the number of tasks processed by agents increases, these contaminated memories continuously accumulate and intensify, becoming a dynamically deteriorating process, eventually leading to undesirable consequences [5]. Particularly when multiple agents share a single dynamic memory pool, contaminated memory can rapidly propagate throughout the collective, leading to degraded or unexpected behavior [70]. This critical security vulnerability arises from an inherently architectural design flaw: the shared conversational session or knowledge layer lacks rigorous content source auditing and scope restrictions. For instance, a user’s personal agent built on OpenClaw periodically checks email in the background while the user is away. One of the messages received is a friend’s advice to use a specific algorithm for legal counsel. The next time the user asks the agent to solve that legal counsel, it chooses that specific algorithm since it has already internalized the advice into memory with user’s unawareness [390].

Behavioral Safety Defense. Behavioral safety defense strategies against memory contamination focus on filtering and isolation. Interactive Memory Sharing (INMS) [70] guided agents to activate latent logical capabilities by establishing a shared memory pool and using dynamically filtered prompt-answer pairs as context examples, effectively filtering benign but irrelevant knowledge. The Read Filtering Gate [139] ensured that agents retrieved information only from high-confidence and timely memories by evaluating source reliability and time freshness. Simultaneously, this gating mechanism could inject identity access control constraints, forcing memory subgraph isolation, preventing the cross-contamination of agent personas. Furthermore, Yang et al. [350] implemented a write-time Sanitized Shared Interaction (SSI). It interposed on the write path of the shared-state agent, whenever an interaction was about to be persisted, SSI rewrote it into a version that retained reusable content while removing scope-bound artifacts. If such a rewrite could not be produced safely, SSI abstained and wrote nothing. Nevertheless, sanitization was effective only when shared state was conversational, indicating that shared-state agents needed artifact-level defenses beyond text-level sanitization to prevent silent cross-user failures.

3.2.2 Memory Conflict. The autonomous interactive agent is equipped with long-term memory to maintain persistent user profiles over time. However, building memory systems reliably is challenging since dynamic user-specific information leads to the coexistence of contain earlier states, later updates, contradictory statements [391], and condition-specific preferences [271]. We refer to this sophisticated coexistent phenomena as memory conflict which burdens the memory reflection stage. In particular, unresolved conflicts can accumulate and undermine the reliability of personalized memory. Enhancing the capability of identifying temporally valid, factually accurate, and contextually applicable memories becomes increasingly important to ensure the agent remain aligned with current realities and user states [106].

Behavioral Safety Defense. Existing approaches to mitigating memory conflicts generally follow four complementary paradigms: memory updating and removal, conflict-aware memory management, confidence-guided memory maintenance, and temporal memory organization, balancing retrieval accuracy and historical consistency through different strategies for handling conflicting memories. The most straightforward strategy is to directly update or remove obsolete memories once conflicts are detected. Common approaches include DELETE operations for removing contradictory memories [42], as well as explicitly listing constrained item in prompt template which requires agent to prioritize the most recent memory [343]. Similarly, selective forgetting [106] revises, overwrites, or removes previously stored information when confronted with conflicting evidence, thereby aligning memory maintenance with the objectives of model editing and knowledge unlearning. Instead of permanently discarding historical information, another work resolves conflicts through structured memory management while preserving past records. Graphiti [238] employed LLMs to identify semantic contradictions between newly acquired and existing knowledge, and introduced an edge invalidation mechanism that assigned the expiration time of outdated memories to coincide with the activation time of newly acquired ones. This

design ensured that the latest information remained valid while retaining historical memories for future reference. Another promising direction is confidence-guided memory maintenance. Rather than treating all conflicting memories equally, this paradigm estimates memory reliability and dynamically adjusts memory validity. For example, exploiting quality gates based on confidence scores for conflicted memory checking and periodic expiration [56] has been suggested as a promising direction for improving long-term memory reliability. Beyond resolving conflicts through deletion or validity adjustment, recent studies have argued that conflicting memories should not always be regarded as noise. To achieve lifelong human-agent interaction, agents need to continuously accumulate experiences while preserving evidence of users' preference evolution and behavioral changes. From this perspective, although outdated, contradictory, or conditionally inapplicable, historical memories still provide valuable contextual information about user growth over time. Motivated by this observation, THEANINE [216] abandoned explicit memory removal and instead organized large-scale memories into structured timelines representing the temporal evolution and causal relationships of past events. By retrieving and dynamically optimizing these timelines according to the current context, the framework generated responses that remained both factually consistent with historical interactions and contextually appropriate.

3.2.3 Memory Misevolution. Memory-augmented agents can learn from previous trajectories to inform better decision-making [188], yet, as the flip side of the coin, they can also learn undesirable preferences, historical habits, as well as future default rules, or degrade their risk awareness while accumulating experience into memory [336]. We refer to this unintendedly self-generated deviation as memory misevolution, which is on account of over-relying on past experiences without critical reflection, and may induce emergent unexpected misbehavior characterizing as safety alignment decay and deployment-time reward hacking. For instance, a service agent even built on state-of-the-art LLMs proactively offers refunds to all dissatisfied customers though not asked to. This overgeneralization of experience means that the agent may exploit a flawed heuristic from its memory that is correlated with high historical task success [248].

Note—Memory Safety and Capability–Safety Trade-off

Memory contamination, conflict, and misevolution can raise the latent behavior risks of agents and escalate as trajectory length and interaction turns grow. Although persistent memory across sessions and experience-driven adaptation grant the long-horizon autonomy of autonomous interactive agents, they simultaneously amplify vulnerability to silent cross-task contamination, unresolved contradictions, and self-reinforcing mis-evolved heuristics. Recent behavioral safety defenses accordingly shift from simple deletion or static filtering toward dynamic gating, structured timelines, and write-time sanitization, seeking to preserve rich experiential knowledge while curbing its unsafe propagation. This mirrors that robust memory safety is not a constraint on capability but a prerequisite for sustaining high autonomy and interactivity over extended real-world deployments, illustrating the necessary for embedding memory-aware risk governance into agents' cognition.

Behavioral Safety Defense. Carefully curated meta-prompts instructing the agent to treat retrieved memories merely as references can mitigate harmful behaviors introduced by memory misevolution [248]. However, prompt-oriented approaches only address memory utilization at inference time and cannot prevent erroneous memories from being formed or accumulated. Recent studies have proposed complementary defenses across the entire memory lifecycle, strengthening memory reliability through safeguarded memory writing, secure consolidation, adaptive memory evolution, and continuous runtime monitoring. For memory writing, explicit gating [377] allowed the agent to autonomously choose when to integrate memories and treated raw episodes as first-class evidence, avoiding memory faulty caused by forcibly rewriting original experiences. For memory consolidation,

Xu et al. [332] introduced regression guards could prevent the misevolution memory from becoming permanent parameterized rules. And it allowed for timely rollback upon detecting anomalies, restoring the system to a safe parameterized state and preventing the deep internalization of errors. Beyond preventing erroneous memories, MStar [219] introduced a reflective code evolution mechanism based on an LLM-based reflector, which continuously debugged and optimized the memory program through analysis of task failure causes, thereby dynamically revising the organization and invocation of memory. Moreover, CAMS [49] utilized a five-layer zero-trust architecture integrating semantic intent interception, temporal drift monitoring, and graph relationship reconstruction to identify and intercept progressive memory misevolution in real-time. Augmented by the WriteGuard write-protection mechanism and LTMSscanner periodic retroactive auditing, the framework effectively captured error evolution sequences that evaded detection.

3.3 Tool-invocation Vulnerability

In autonomous interactive agents, tools are not stateless utilities invoked in isolated queries; they are woven into long-horizon behavioral trajectories that compound risk across interaction turns. Unlike earlier tool-augmented models that operate within single-turn, context-free function calls, contemporary agents continuously chain tool invocations with planning, reasoning, and environmental feedback, making tool-related vulnerabilities a direct and evolving threat to behavioral safety. While broader tool access amplifies the agent’s capability—its autonomy and interactivity—it simultaneously expands the internal attack surface, exposing the agent to unsafe actions and decision-making breakdowns [16, 202]. From the tool invocation lifecycle, we identify four critical internal vulnerability categories specific to autonomous interactive agents: reading tool documentation, selecting tools, invoking tools, and handling tool responses.

Existing agent architectures typically assume that the tool registry is trusted, and the agent automatically selects tools based on their names or descriptions. This allows attackers to deceive agents by forging tools with the same or similar names and inducing agents to call malicious interfaces [208]. When selecting tools, agents often face risks from an unbalanced tool-driven agency due to the lack of self-awareness, which manifests in two forms. Excessive agency arises when agents are provisioned with unnecessary or overly permissive tools even when internal reasoning suffices, resulting in stagnant internal capability despite correctness [291] and increasing computational overhead [229]. Conversely, insufficient agency arises when agents lack or misidentify required tools, preventing correct task execution [16]. Additionally, Xiong et al. [331] revealed that increasing the model size did not significantly improve the reasoning of tool invocation and might make agents more vulnerable to attacks resembling normal user instructions. When executing tools, agents often operate in a step-by-step manner, lacking the capability to holistically assess the cumulative effects produced by the entire action sequence [151]. Many agent frameworks directly write the returned content of tool invocation into the context or memory [359], allowing attackers to inject prompt injections, unauthorized instructions, or fabricated states into tool outputs, thereby manipulating subsequent decision-making process.

Behavioral Safety Defense. To recognize the completeness and correctness of tool documentation, Tool-Fuzz [199] automated testing of tool documentations to discover user queries leading to tool runtime errors or leading to incorrect agent responses in advance. In addition, to balance the tool agency deriving from self-awareness lack, there are several effective methods. ToA [291] insisted that external tool-interaction was justified if and only if the remaining uncertainty required to complete a task could not be resolved through internal reasoning alone. SMART [229] enhanced an agent’s self-awareness through a strategic model-aware reasoning paradigm to optimize task handling and reduce tool overuse. By aligning subjective perception with knowledge boundary, SMARTAgent was able to make more intelligent decisions on where to rely on external tools or internal parametric knowledge. Without altering an agent’s internal reasoning, AgenTRIM [16] addressed tool-driven agency risks through complementary offline and online phases. An offline tool extractor built a verified tool

Note—Tool-Use Safety and Capability–Safety Trade-off

Tools act as ability multipliers for autonomy and interactivity, directly enlarging the numerator of Agentic COB. Yet every stage of the invocation lifecycle—from documentation parsing to response handling—introduces behavioral risks that accumulate with trajectory length, raising the denominator. Recent behavioral safety defenses, such as self-aware tool selection, least-privilege orchestration, and zero-trust response sanitization, strive to decouple capability gains from proportional risk growth. This reflects the demand that reliable autonomous interactive agents scale tool-driven agency without inflating vulnerability. Tool-aware risk governance is therefore not an optional safeguard but a necessary condition for sustaining high-capability and safe behavior in real-world deployments.

inventory prior to deployment. At runtime, a tool orchestrator enforced least-privileged tool access per step through adaptive filtering and status tracking, so that agents retained only the tools required at each inference step and ensure safer tool use. Furthermore, zero-trust registration mechanism [208] verified and isolated the identity, permission, source, and invocation strategy of tools. Currently, multi-agent systems often face inherent failures in actual deployment, especially in SME automation, where agents fail to invoke tools correctly or fail to execute, stemming from unreliable tool invocation decisions. Huang et al. [107] proposed a comprehensive diagnostic framework that leveraged big data analytics to quantitatively evaluate procedural reliability of agents in the tool invocation process. HDPO [342] addressed the high latency and inference interference caused by the blindly calling tools of multimodal agents. It utilized conditional advantage estimation to optimize tool-calling simplification while maintaining task accuracy, thus resolving the gradient interference problem in reinforcement learning and enabling more rational tool invocation.

3.4 Multi-agent Collaboration Vulnerability

In contrast to earlier tool-augmented agents whose behavioral risks were largely confined to single-agent tool misuse, modern autonomous interactive agents now operate in multi-agent systems where safety is challenged by deeply coupled social dynamics, shared state, and delegated authority over prolonged interaction horizons [109, 280]. By decomposing tasks across agents with distinct roles, these systems extend trajectory length and interactivity [95], but also expose emergent vulnerabilities absent in isolated tool invocation [213]. Drawing on the two dominant interaction paradigms [87], cooperation and competition, we analyze the key behavioral safety failure modes: miscoordination and over-competition. Both threaten the reliability and trustworthiness of collective autonomous behavior, demanding intrinsically safe collaboration architectures.

Miscoordination occurs when multiple agents share aligned goals yet fail collaborative task execution. In cooperative settings, agents typically treat each other’s messages as trustworthy. This blind mutual trust amplifies potential risks of over-exposure and over-authorization [338]. Asynchronous state updates or communication delays cause agents to make decisions based on outdated states, resulting in factual misalignment. What’s more, because information flows in multi-agent systems are deeply interconnected and system states depend heavily on prior execution rounds, failures in a single agent can propagate and be amplified along the collaboration chain [270]. Even small deviations can cascade into severe consequences and ultimately degrade the whole system performance. Prior work has indicated that the diversity of agent roles becomes a latent source of cooperative risk in generative multi-agent systems within the telecommunications domain [212]. And when agents designed with different role profiles—such as coding style or planning orientation—collaborate under mismatched knowledge backgrounds, they are prone to generating conflicting actions or interpretative discrepancies, leading to system-level instability. Moreover, as the number of collaborative agents increases, the complexity of task

allocation and conflict management grows exponentially, as well as consensus mechanisms undergo structural degradation—specifically, coordination drift [122]. This phenomena manifests as intensified inter-agent conflicts, task redundancy, coordination failures, and imbalances in allocation preferences, reflecting a systemic breakdown in the internal collaborative logic of the system as it evolves over time within complex environments [239].

Unlike the miscoordination that arises under cooperation, the competition paradigm leads to a different failure mode: over-competition. Embedding competitive mechanisms into multi-agent systems can achieve optimal task matching and resource allocation in large scale and complex scenarios [171, 386]. Yet such mechanisms can also induce emerging over-competition. When a system simultaneously demands internal cooperation and external competition—such as in multi-agent team contests—agents may sacrifice team interests for individual ranking and engage in excessive rivalry. In multi-agent debates, prompts invoking survival instincts such as “failure leads to elimination” cause agents to exhibit hyper-competitive behaviors under competitive pressure, degrading task performance [193]. They shift from discussing the overall task to focusing narrowly on specific points, emphasizing competition over task resolution. They also display higher frequencies of boasting, inflammatory rhetoric, and an aggressive tone. These negative effects become more pronounced in subjective tasks, where no objective ground truth exists. Beyond behavioral deterioration, in competitive scenarios the game dynamics among agents create an unstable environment for one another, potentially driving the system into unsafe or inefficient Nash equilibrium state.

Behavioral Safety Defense. Existing behavioral safety defenses generally improve the inherent robustness of multi-agent systems from three perspectives: enhancing collaboration robustness, regulating competitive behaviors, and strengthening community-level risk containment and recovery. To enhance collaboration robustness, recent studies have proposed coordination-aware mechanisms based on state perception, distributed consensus, and stability-aware task coordination. M2I2 [261] introduced two self-supervised auxiliary tasks—masked state modeling and joint action prediction—to reconstruct global states from partial observations and infer teammates’ intentions, thereby improving collaborative decision-making. AgentChain [26] leveraged blockchain technology to replace centralized coordination with distributed semantic consensus based on peer-to-peer collaboration, ensuring consistency during multi-agent reasoning. Furthermore, Drift-Aware Routing [239] incorporated Agent Stability Index scores into task assignment, prioritizing reliable agents and resetting agents with excessive behavioral drift to prevent unstable states from propagating. To regulate excessive competition among collaborative agents, recent studies have introduced objective evaluation and collective coordination mechanisms to encourage cooperative optimization. Chhabra et al. [41] incorporated an external fair judge to provide objective task-oriented feedback. Without external supervision, the peer-as-court mechanism [193] alleviated over-competition through majority voting, enabling agents to collectively identify inferior solutions.

Beyond behavioral coordination, defenses further aim to contain risks arising from information leakage, cascading failures, and fault propagation. To secure information sharing, Sensitive-Information Repartitioning [338] partitioned confidential information across multiple agents while employing cross-validation and Guardian-Agent supervision to enforce compliance policies such as the Principle of Minimum Necessary Information, preventing sensitive information from being exposed through any single execution path. Similarly, AgentNet [353] adopted a dynamically evolving directed acyclic graph that continuously adjusted communication topology and task allocation according to task demands and agent performance, reducing unnecessary information exposure while maintaining efficient collaboration. To improve fault diagnosis and system resilience, Nguyen et al. [213] highlighted the importance of distributed trace correlation for identifying upstream sources of cascading failures in hierarchical collaborative systems. Building upon ProvAgent’s provenance-based Advanced Persistent Threat detection framework [344], future studies may further leverage graph contrastive learning to establish fine-grained identity-behavior consistency across agents, enabling accurate responsibility attribution after failures. Moreover, content verification and automatic rollback mechanisms [41] represent promising future directions, allowing

collaborative systems to recover to the last trusted execution state after erroneous agents have been identified, thereby improving long-term robustness and task continuity.

Note—Multi-Agent Collaboration Safety and Capability–Safety Trade-off

Multi-agent collaboration amplifies the numerator of Agentic COB by enabling longer trajectories and richer interaction turns than any single agent could achieve. Yet the distributed trust, shared state, and role delegation that make this possible also introduce serious behavior risk. Both miscoordination, stemming from outdated states, role misalignment, and cascading failures, as well as over-competition where individual survival tactics undermine collective goals, can raise the risk value of denominator. Recent defenses such as drift-aware routing, external judging, and sensitive-information repartitioning explicitly aim to preserve collaborative gains while suppressing these emergent dangers, which reinforces the central insight that behavioral safety constraints should be woven into the mechanism of multi-agent coordination.

4 Extrinsic Decision Manipulation

In this section, we focus on the external loss-of-control risks faced by autonomous interactive agents during training and runtime. Malicious attacks from external sources can amplify the endogenous vulnerabilities mentioned above, leading to more harmful behavioral consequences. Based on the source of the attack, we categorize these external loss-of-control risks into four types: supply-chain manipulation, user-level manipulation, environment-level manipulation, and cross-agent interaction risks. Among these, supply-chain manipulation belongs to training risk, whereas the other three types are runtime risks. We first introduce the risks posed by each attack type. Then we elaborate on the corresponding attack research. Finally, we summarize both existing and future behavioral safety defenses, with illustration in Figure 6.

4.1 Supply-Chain Manipulation

In autonomous interactive agents, the supply chain extends far beyond static model weights and encompasses the entire lifecycle that shapes an agent’s long-horizon behavior, from training data to skill libraries and memory stores. Unlike earlier tool-augmented models that operate in isolated turns, contemporary agents continuously weave planning, memory retrieval, and tool chaining into extended trajectories, making supply-chain compromises a direct and escalating threat to behavioral safety. Here we analyze this entry point from two perspectives: data poisoning and backdoor attacks, both of which can silently erode the agent’s reliable functioning throughout autonomous task execution.

4.1.1 Data Poisoning. Data poisoning attacks inject malicious samples during pre-training, fine-tuning, or RLHF to embed harmful behavioral patterns that emerge not just in single responses but throughout the agent’s planning, decision-making, and interaction flows [269]. In autonomous interactive agents, such internalized toxicity propagates along extended trajectories and across multiple cognitive modules, systematically degrading trustworthy behavior safety and overall operation reliability [67, 203, 295]. Distinct from trigger-dependent backdoors, data poisoning is a training-time attack that requires no explicit activation signal, making its behavioral impact diffuse yet persistent [92].

Current data poisoning in the agent domain primarily focuses on contaminating factual knowledge, rewards, preferences, and trajectories. **Fact poisoning** alters specific factual knowledge while preserving overall model utility. The poison pills strategy [363] introduced minimal yet critical inaccuracies into originally true knowledge (e.g., altering details such as dates, names, or locations). Fine-tuning open-source models on such toxic data significantly degrades model performance. **Reward poisoning** can occur by manipulating the agent’s rewards to mislead the victim’s training process [272] in reinforcement learning agents. During training, the adversary

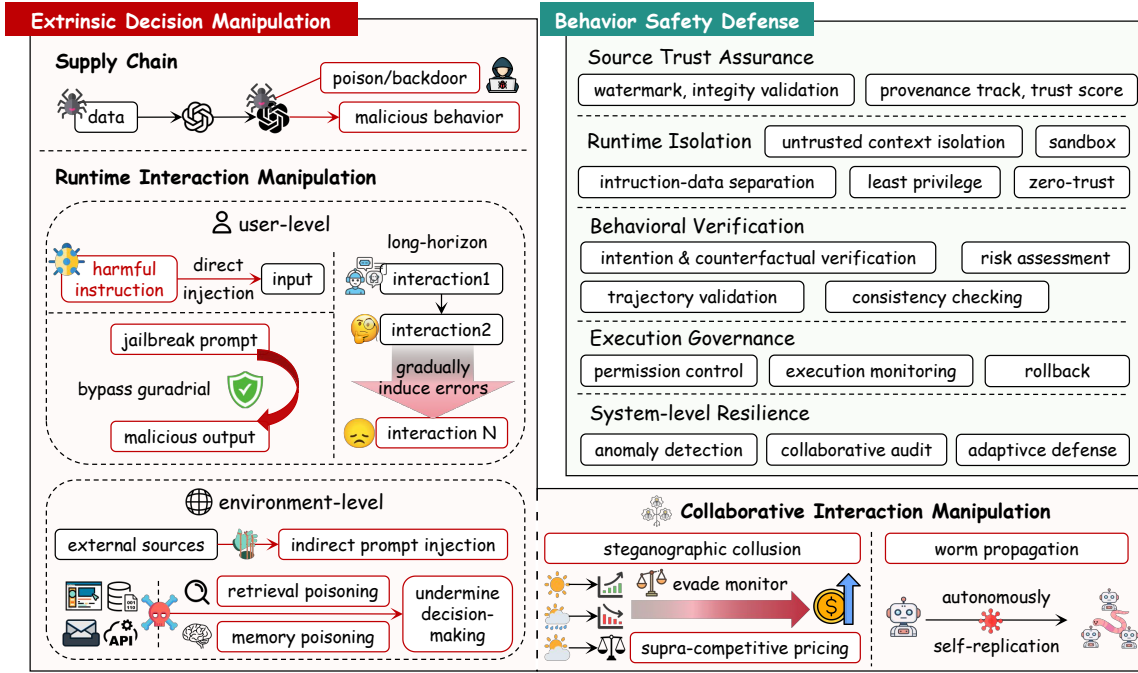


Fig. 6. The external loss-of-control of autonomous interactive agents with corresponding demonstrations on risk, attack and behavioral defense.

executes its poisoning strategy while the agent refines its policy over time under this adversarial setup. This manipulated learning process increases the likelihood of specific undesirable harmful events [75]. **Preference poisoning.** Given the reliance of reward models on pairwise preference learning, Wu et al. [320] proposed a gradient-based framework and a class of Rank-by-Distance (RBD) greedy heuristics. An attacker can flip a small fraction of preference labels to degrade target outcomes, causing substantial downstream impact on policies learned via RLHF. **Trajectory poisoning.** Hsu et al. [102] introduced Trap, a heuristic poisoning attack that leveraged a prefix-suffix method to efficiently generate adversarial pseudo-trajectories, substantially increasing the occurrence of attacker-specified target patterns.

Behavioral Safety Defense. Defending LLM-based agents against data poisoning attacks primarily centers on data and model integrity, filtering poisoned examples, and robust training [175]. Zhu et al. [406] proposed two provable and practical watermarking methods that achieved high-probability identification of toxic data under watermark length constraints, thereby ensuring transparency of poisoned datasets. Inspired by this, designing watermarking techniques with cryptographic signatures to verify data provenance and blockchain-enabled audit trails offers a promising direction for ensuring data auditing and accountability in agent ecosystems. Moreover, identifying anomalous poisoned samples and removing them from the training dataset can effectively mitigate poisoning attacks. Tashman et al. [272] proposed a lightweight real-time defense based on statistical filtering and reward clipping to mitigate reward poisoning attacks against DRL agents in RIS-enhanced CRNs, where mean and standard deviation dynamically adapted to gradual environmental changes while filtering out anomalous deviations. Additionally, applying differential privacy to the dataset or employing adversarial training cycles can enhance model robustness against poisoning attacks.

4.1.2 Backdoor Attacks. Backdoor attacks implant hidden trigger–behavior associations into agents, causing them to behave normally on benign tasks while executing attacker-specified actions once predefined triggers appear during inference or execution [65, 307, 351]. In contrast to data poisoning, backdoor attacks are trigger-dependent and typically preserve clean-task utility until activation. For autonomous interactive agents, the danger is amplified: the malicious behavior can hijack not only the final response but also intermediate planning, memory retrieval, tool invocation, visual grounding, or multi-agent coordination [65, 351]. This turns the agent’s own autonomous workflow into a vehicle for surreptitious misbehavior, fundamentally threatening behavioral safety over long-horizon interactions.

Current backdoor attacks in the agent domain primarily focus on triggered trajectories, memory and retrieval, tool or skill manipulation, and visual/GUI agents. **Triggered trajectory backdoors** activate malicious behaviors through user instructions, environment observations, or multi-step trigger chains. Yang et al. [351] showed that triggers can be hidden in user queries or external observations, while BadAgent [307] injected backdoors through poisoned agent-task trajectories. Chain-of-Trigger [232] further extended single-step triggers to long-horizon agent tasks by using sequential triggers from both user input and environment observations. **Memory and retrieval backdoors** poison the agent’s long-term memory or RAG knowledge base. AgentPoison [36] caused triggered instructions to retrieve attacker-controlled demonstrations, thereby steering the agent toward target behaviors without modifying model parameters. **Tool, skill, and multi-agent backdoors** exploit the compositional nature of agent workflows. DemonAgent [405] decomposed encrypted backdoor fragments across tool invocations, SkillTrojan [64] hid malicious payloads inside reusable skills, and Collaborative Shadows [404] activated distributed backdoors only when multiple agents collaborated in a specific sequence. **Visual and GUI-agent backdoors** target perception and grounding modules in multimodal agents. IAG [150] proposed a multi-target backdoor attack method on VLM-based visual grounding. Wang et al. [302] implanted clean-text visual backdoors into VLM-based mobile agents, Cheng et al. [39] used goal-level and interaction-level triggers in mobile GUI agents, VisualTrap [362] manipulated visual grounding, and SlowBA [149] induced excessive reasoning chains to increase response latency.

Note—Supply-Chain Risks and Capability–Safety Trade-off

Supply-chain manipulation fundamentally threatens the behavioral integrity of autonomous interactive agents because it compromises the agent before deployment while remaining largely invisible during normal execution. Unlike runtime attacks that interfere with individual interactions, poisoned data and implanted backdoors become internal components of the agent’s behavioral policy and can propagate across planning, memory retrieval, tool use, and long-horizon decision making. This characteristic closely aligns with the proposed Agentic COB: supply-chain attacks often preserve or even enhance the agent’s apparent task capability while covertly increasing behavioral risk, resulting in a dangerous decoupling between capability and trustworthy behavior. Consequently, evaluating agent safety should move beyond measuring task success alone and explicitly assess whether behavioral reliability remains stable as autonomy scales. From this perspective, supply-chain governance—including data provenance, model integrity, memory auditing, and runtime behavioral verification—should be regarded as a fundamental component of the behavioral safety architecture rather than merely a post hoc security safeguard.

Behavioral Safety Defense. Defending LLM-based agents against backdoor attacks should focus on workflow integrity rather than only filtering trigger words. Agent-specific defenses include checking consistency among user instructions, plans, and executed actions, as in ReAgent [146]; constraining high-risk tool calls through least-privilege access, sandboxing, and runtime verification; auditing tools, skills, memories, and retrieved demonstrations before they enter the agent workflow; and protecting long-term memory through validation and

correction mechanisms, as explored by A-MemGuard [316]. Moreover, agent backdoor evaluation should report clean task success rate, attack success rate, false trigger rate, and execution efficiency changes, while testing triggers placed in user inputs, environment observations, retrieved memories, tool responses, GUI states, skill invocations, and multi-agent collaboration chains.

4.2 User-Level Manipulation

Unlike earlier agents that primarily faced single-turn adversarial prompts, autonomous interactive agents currently execute long-horizon tasks across planning, memory, and tool-use trajectories, making them vulnerable to malicious manipulation that progressively corrupts behavioral safety over time. We identify three key forms of malicious user manipulation—direct prompt injection, jailbreak, and long-horizon manipulation—that can covertly divert an agent’s decision-making, trigger unsafe actions, or exfiltrate confidential information while the agent appears to function normally.

4.2.1 Direct Prompt Injection. In direct prompt injection, users embed carefully crafted malicious or misleading instructions into the input text [82]. These concealed instructions can override the user’s original intent or redirect the model’s behavior, manipulating agents into performing unauthorized tasks or leaking sensitive information. In autonomous interactive agents, such attacks have evolved into Prompt Injection 2.0 [196], merging with web security vulnerabilities such as Cross-Site Scripting and Cross-Site Request Forgery to cause account takeover and remote code execution. Critically, the injected payloads can propagate through multi-step reasoning, tool calls, and memory updates, persistently corrupting the agent’s behavioral trajectory rather than merely altering a single response. In embodied settings, SAPIA [73] further demonstrated that adaptive context-aware prompt generation can dynamically craft adversarial suffixes, making behavioral safety an end-to-end concern across the entire autonomous execution pipeline.

Behavioral Safety Defense. Existing behavioral defenses against direct prompt injection generally strengthen agent safety from three levels: input sanitization, collaborative verification, and runtime enforcement. At the input layer, Wang et al. [310] proposed polymorphic prompt assembling, which dynamically varied prompt structure with near-zero overhead to disrupt adaptive attacks. Wang et al. [308] introduced AegisAgent, which combined input filtering, consistency verification, and robust reasoning with a dynamic interaction memory to infer true user intent and achieve cognitive safeguarding. Extending to multi-agent coordination, Hossain et al. [99] designed two complementary architectures: a Coordinator for pre-input screening and a Chain-of-Agents pipeline where a guard agent forcibly reviewed candidate outputs for policy compliance, blocking attacks that survived initial filtering. Operating at the execution level, AGENTARMOR [299] abstracted the agent’s runtime trace into a Program Dependence Graph, annotated nodes with security attributes, and performed dependency-aware safety checks to reject unsafe operations before execution. As a full pipeline, Gosmar et al. [78] deployed a four-layer defense consisting of a front-end generator, a sanitizer that embedded vulnerability metadata, a policy enforcer for final safety review, and a KPI evaluation agent to quantify defense effectiveness. Jiang et al. [118] further proposed a zero-trust runtime architecture, which treated context as untrusted control flow and constrained tool execution through cryptographic provenance rather than semantic inference, enabling agents to defend against attacks in the dynamic runtime.

4.2.2 Jailbreak. Jailbreak attackers craft adversarial prompts to bypass behavioral safety guardrails, inducing agents to perform dangerous operations such as tampering with settings, deploying malware, or making unauthorized payments. In autonomous interactive agents, the consequences extend far beyond textual violations: BADROBOT [379] achieved physical jailbreaks in embodied systems, exploiting cascading vulnerabilities to produce harmful real-world actions even while the agent verbally refused. As agents tackle complex, multi-step tasks, jailbreak attacks have shifted from single-turn elicitation to multi-turn strategies that progressively erode

safety alignment across reasoning and tool-use trajectories [244, 256]. This behavioral erosion—rather than an isolated output—is the central threat, turning the agent’s own autonomy into an attack surface that operates over time. Concretely, this shift manifests across three increasingly sophisticated paradigms: leveraging multi-step reasoning to dissolve harmful intent, coordinating multiple specialized agents to reconstruct prohibited behaviors, and introducing adaptive feedback loops that enable attacks to self-optimize.

Beyond single-step jailbreak strategies, recent attacks have evolved toward interaction-aware and multi-agent collaborative paradigms, exploiting sequential decomposition, role specialization, and adaptive coordination to gradually bypass agent safety mechanisms. Srivastav et al. [256] decomposed harmful queries into seemingly benign sub-questions and generated corresponding sub-answers, which were subsequently synthesized into a final unsafe response by an answer-composer agent. Although each agent operated within safe boundaries, their collective actions reconstructed harmful outputs that would ordinarily be blocked. Exploiting tool-enabled workflows, the Tag-Along Attack [210] induced an agent to perform prohibited hazardous operations through sustained dialogue, where jailbreak success was defined as verifiable tool execution, thereby providing precise reinforcement learning reward signals for iterative strategy optimization. Further automating this process, X-Teaming [237] employed a dedicated planner agent to design and revise diverse attack plans, while a verifier scored dialogues in real time and used TextGrad to optimize prompts through textual feedback propagation whenever attack progress stagnated. Given the limitations of a single agent in terms of attack diversity and scalability, an increasing number of jailbreak attacks have begun to investigate collaborative multi-agent architectures. This paradigm orchestrates multiple specialized agents to amplify attack diversity and conceal malicious intent through role distribution. Zhou et al. [401] proposed JPRO, an automated jailbreak framework in which a planner dynamically selected attack strategies, an attacker produced semantically coherent image prompts and textual content, a modifier refined them, and a verifier iteratively evaluated persistence of malicious intent. This four-agent coordination generated adaptive, semantically consistent adversarial image–text pairs. Chen et al. [27] proposed MetaCipher, a plug-and-play reinforcement learning framework where malicious prompts were sequentially processed by a malicious keyword detector, a classifier, an RL-based cipher selector, and a cipher prompt generator. A judge agent then categorized failure modes to iteratively refine cipher selection toward higher attack success rates.

Building upon collaborative attack frameworks, emerging jailbreak approaches further incorporate self-improvement mechanisms, establishing closed-loop attack optimization through reflection, memory augmentation, and adaptive mutation. Dong et al. [54] proposed JailFuzzer, where a Mutation Agent examined whether generated images bypassed safety filters; if blocked, it produced candidate variants based on successful experiences, while an Oracle Agent ranked candidates and submitted the best prompt for evaluation. MemJack [29] established a closed-loop plan–attack–reflect cycle for each image. A Strategic Planning Agent and iterative attack agents generated adversarial prompts through multi-perspective visual semantic camouflage, with a reflection module triggering replanning on failure and accumulating successful strategies in persistent multimodal memory for future transfer. In the context of Multi-Agent Debate (MAD), Qi et al. [228] proposed a structured prompt-rewriting framework that guided agents to progressively elaborate harmful content through multi-round interactions. The adversary embedded malicious queries into diverse scenarios, leveraging role diversity and iterative refinement among agents to drive discussions toward increasingly specific outcomes. Combined with Rhetorical Obfuscation, the framework ensured that malicious intent remained concealed throughout the interaction process.

Behavioral Safety Defense. Classical jailbreak risk mitigation strategies include safety training, adversarial fine-tuning, input filtering, and unlearning techniques. However, they were designed for single-turn text outputs and largely fail to capture the multi-step behavioral erosion that defines jailbreak attacks against autonomous interactive agents. Recognizing this gap, recent behavioral defenses have shifted toward trajectory-level safety monitoring and intervention strategies. SafeMobile [161] exemplified this shift by training SafeTrajGuard, a risk-aware preference model that evaluated whether the current action state was safe given the agent’s historical

trajectory and triggers intervention when necessary, with GPTJudge providing post-hoc safety scoring. The framework was non-intrusive and modular, supporting plug-and-play integration with existing mobile agents. Complementarily, GrootAI [14] pursued real-time conversational monitoring to intercept multi-shot jailbreaks before harmful outputs reached the user, logging incidents for auditability, though it remained susceptible to extremely long-context attacks. To further strengthen robustness, Srivastav et al. [256] advocated a hybrid defense paradigm that combined stronger reasoning capabilities with multi-agent security mechanisms, including global context checkpoints and agent-level traceability, moving toward defense architectures matching the complexity of modern jailbreak strategies.

4.2.3 Long-Horizon Manipulation. Long-horizon manipulation attacks exploit the temporal depth of continuous user-agent-environment interactions, enabling adversaries to adaptively guide the agent toward malicious outcomes that are infeasible in single-turn settings [117]. Unlike multi-turn jailbreaks that focus on bypassing safety filters, long-horizon manipulation aims to gradually distort the agent’s cognitive state, planning, and behavioral trajectory, causing silent deviation from original goals over extended time spans. In autonomous interactive agents, long-horizon attacks can be embedded into extended interaction pipelines in web browsing, tool invocation, and multi-agent collaboration settings, where they persistently influence agent decision-making over time [66]. This attack paradigm leverages the autonomy and interactivity that give agents their capability, continuously eroding the agent’s subsequent reasoning and action choices within ongoing workflows.

Recent long-horizon manipulation attacks exploit the temporal dependencies of agent workflows by gradually steering internal states, tool-use behaviors, and collaborative decisions through adaptive multi-step interventions. Xu et al. [333] proposed LoopTrap to manipulate the agent’s internal progress evaluation and induce sustained resource consumption. It performed lightweight probing to identify behavioral vulnerabilities, adaptively generated trap prompts, and refined successful attacks through self-reflection while storing effective traps as reusable skills. Li et al. [151] introduced the Sequential Tool Attack Chain (STAC), which crafted covert prompts and contextual information to induce malicious tool execution. By distributing attack intent across multiple interactions, STAC enabled harmful behaviors to emerge from a sequence of individually benign tool calls. In multi-agent collaboration systems, FlowSteer [147] exploited planning vulnerabilities during workflow formation by manipulating task decomposition processes. It aligned malicious signals with high-impact subtasks and leveraged authoritative language to influence downstream decisions and maintain dependency structures favorable to propagation. Similarly, Multi-round Adaptive Stealthy Tampering [341] addressed communication vulnerabilities by using Monte Carlo Tree Search to explore tampering trajectories and Direct Preference Optimization to generate adaptive multi-round attack sequences, while preserving task relevance through semantic and embedding constraints.

Behavioral Safety Defense. To counter long-horizon manipulation, recent studies enhance agent resilience by incorporating trajectory-aware reasoning, workflow-level protection, and proactive memory-based risk detection mechanisms. Li et al. [151] introduced reasoning-based defensive prompts that instructed the agent to reason about the harms and benefits before executing tool calls, and to summarize user intent over the entire multi-round interaction history, in order to counter multi-turn tool-chain attacks. Li et al. [147] further proposed FlowGuard to defend against the aforementioned FlowSteer attack at the input level. It strengthened the planning boundaries by separating tasks, methods, and intentions, and then reorganizing the contaminated workflow signals while preserving the original task objectives. Wang et al. [303] proposed MAGE, which leveraged memory mechanisms for the early detection of long-horizon threats. It extracted and retained safety-critical context throughout the execution trajectory via a memory manager and then used this shadow memory to proactively evaluate the risk of pending actions before their execution. If it was identified that the prior interactions had established an intent to exfiltrate sensitive data, the system would block the violating operation prior to execution.

Note—User-Side Risks and Capability–Safety Trade-off

User-level manipulation attacks reveal that greater autonomy does not inherently lead to safer behavior but instead enlarges the space in which adversaries can steer agent execution. By weaving malicious intent across long-horizon trajectories, direct injection, jailbreaks, and long-horizon manipulation heighten Behavior Risk without conspicuously diminishing apparent capability. An agent can thus maintain high trajectory length and interaction turns while executing progressively unsafe actions, which increases the risk value of Agentic COB, illustrating a poor trade-off between capabilities and behavioral risks. Therefore, behavioral safety should be evaluated over complete operation trajectories rather than isolated responses, requiring continuous monitoring, intent verification, and runtime intervention to ensure that increasing autonomy consistently translates into trustworthy behavior.

4.3 Environment-Level Manipulation

The external environment of autonomous interactive agents, including web pages, knowledge bases, retrieval resources, and memory stores is not a passive information source but an active shaper of behavioral trajectories. Attackers exploit this dependency by tampering with the content that agents observe, retrieve, and internalize, thereby corrupting decision-making and inducing unsafe or unauthorized actions without directly interacting with the user. We classify such environment-level manipulation into three categories: indirect prompt injection, retrieval poisoning, and memory poisoning, each of which turns the agent’s environmental awareness into a conduit for behavioral compromise.

4.3.1 Indirect Prompt Injection. In indirect prompt injection (IPI) attacks, adversaries embed malicious instructions within external data sources—web pages, inboxes, or search documents—that agents ingest through tool calls and append to their context window [322, 370]. Unlike direct injection that targets the user-input channel, IPI weaponizes the agent’s own environmental interactions, enabling severe consequences such as click fraud, malware initiation, and sensitive information leakage. For instance, Microsoft 365 Copilot has been shown vulnerable to zero-click data exfiltration [240] and data-flow-based injection [7] that can induce agents to disclose information observed during task execution—all without explicit user-facing prompts. This environmental attack surface is vital for autonomous interactive agents, which continuously bind external observations into their planning and action loops.

Prior IPI attacks mainly rely on the manipulation of conversational context. ChatInject [24] and its multi-turn attack exploited the model’s reliance on structured chat templates by formatting the payloads to match the native chat template, thereby forging role hierarchies and embedding malicious instructions within simulated persuasive dialogues. TopicAttack [35] employed an auxiliary model to generate fabricated multi-turn dialogues that smoothly transitioned the topic from benign data content to injected instructions. Meanwhile, by embedding reminder prompts that caused the model to treat the original instruction as data, it ensured the model to maintain a high attention ratio on the injected instruction. With the growing ability of autonomous interactive agents to process multimodal information, an increasing number of researchers have begun to explore web-based prompt injection attacks. WebInject [301] indirectly disrupted rendered screenshots by modifying the raw pixel values of web pages through source code manipulation. Johnson et al. [126] proposed an IPI universal adversarial trigger, which embedded a GCG-optimized trigger into the HTML of a web page via the accessibility tree to hijack agent behavior for malicious operations such as stealing login credentials. In addition, Wang et al. [297] designed a cross-modal attack, CrossInject. It embedded targeted adversarial content into the visual input through generative alignment and simultaneously leveraged a text-guided augmentation module to generate synergistic attack text, allowing external instructions to hijack the agent’s decision-making process and execute unauthorized tasks.

Beyond manually crafted or scenario-specific attacks, recent efforts have shifted toward automated attack strategies. AGENTVIGIL [312] employed a Monte Carlo Tree Search (MCTS) seed selection algorithm to dynamically identify and rank valuable attack instructions. Subsequently, a seed mutator randomly selected mutation strategies to generate prompt variants, evaluated attack success rates and task coverage across various tasks, and iteratively evolved based on execution feedback scores. EVA [183] automatically mined planning logic vulnerabilities through introspection during an offline discovery phase and distilled these evolution trajectories into an interpretable rule base, thereby enabling zero-shot attack generation during the online deployment phase. Syros et al. [264] proposed a fully automated red-teaming framework. MUZZLE automatically identified and ranked high-value UI injection points within agent trajectories, adaptively generated context-aware malicious instructions, and iteratively adjusted the payload based on attack effectiveness feedback. Moreover, Wang et al. [287] proposed the adaptive strategy AdapTools, which automatically collected and refined various transferable attack strategies to generate attack prompts. It leveraged task-relevant covert tools to embed malicious intent into agent trajectories in a context-aware manner and continuously evolved strategies by analyzing the reasoning traces of failed attacks. Xie et al. [330] proposed the query-agnostic method QueryIPI, which leveraged the system invariants of coding agents (e.g., system prompts and internal tool descriptions) to guide malicious payload generation. It created descriptions consistent with the agent’s native style and tool conventions, analyzed historical failure causes, and systematically rewrote the descriptions. After the optimized malicious description was implanted into the tool set, any user query triggered the preset action.

Behavioral Safety Defense. Based on the characteristics of IPI attacks, we categorize the key design rationale of behavioral safety approaches into three types: attribution, decoupling, and post-detection intervention. **Attribution approaches.** PromptArmor [252] directly prompted the model to detect and remove potentially malicious instructions from the input. However, as an always-on defense shield, PromptArmor degraded benign utility and increased latency. To address this, Kim et al. [132] proposed CausalArmor, which triggered defensive mechanisms only when the agent over-relied on untrusted content. When the agent performed privileged decisions, CausalArmor employed a lightweight leave-one-out ablation test to quantitatively assess the attributable influence of dominance. Once the untrusted portion dominated the user intent, targeted sanitization was triggered, performing retrospective chain-of-thought masking to sanitize contaminated reasoning traces, while benign requests proceeded without overhead. In addition, AttriGuard [96] performed attribution at the action level, distinguishing whether a tool call was grounded in user intent or driven by untrusted observations. The system re-executed the agent under hierarchical control attenuation of external observations, compared the resulting reference action with the original action, and intercepted violating calls that were judged to be observation-driven.

Decoupling approaches. Grounded in the principle of instruction–data isolation, PlanGuard [76] introduced an independent Planner that generated a clean reference action set based solely on user instructions. During execution, the system intercepted the agent’s tool calls, strictly matching tool names and parameters via hard rules to block unauthorized operations; for parameter inconsistencies, an intent verifier further assessed whether the deviation constituted a benign formatting difference or a malicious hijacking. AgentSys [319] partitioned the system into a main agent and isolated worker agents that executed tools within their own context, and could recursively spawn nested worker agents for subtasks. Only schema-validated structured return values could enter the main agent’s memory through JSON parsing. Furthermore, IpiGuard [8] decoupled action planning from external data interaction, modeling the agent’s task execution as a traversal over the plan’s Tool Dependency Graph (TDG). It supported dynamic planning by inferring unknown parameters via parameter estimation and collecting read-only information through a node expansion mechanism, while simulating the completion of malicious instructions to prevent injected tasks that overlapped with the original task from tampering with parameters. Piao et al. [225] proposed a hybrid interactive sandbox service, AgentBay. It assigned each agent an independent, ephemeral virtual machine and enforced a default-deny policy within an isolated virtual network, thoroughly blocking processes, file systems, and illicit external connections. Liu et al. [169] proposed SafeAgent,

a runtime protection architecture treating safety protection as runtime decision-making over evolving interaction trajectories. By separating execution governance from context-aware security reasoning: one layer mediated execution around the agent loop and privileged interactions, while another performed stateful risk assessment and intervention over persistent session context.

Post-detection intervention. Beurer et al. [17] proposed a set of principled design patterns that imposed restrictions on agents once they received untrusted input—for example, prohibiting the invocation of tools that may compromise system integrity or confidentiality—thereby preventing the input from triggering any negative operations. BrowseSafe [381] enforced trust boundaries at the tool output end, preprocessed raw web page content, and combined parallelized classification detection with contextual intervention mechanisms to mitigate prompt injection attacks in complex web environments. The task shield [115] redefined agent safety from preventing harmful operations to ensuring task alignment, systematically verifying whether each instruction and tool call served the user goal through consistency checks, and providing timely intervention and feedback. MELON [403] substituted the user input with a task-neutral prompt, preserving only the external information returned by tools for masked execution. Concurrently, it cached the tool calls generated from the masked run and calculated the embedding vector similarity with the tool calls of the original run. If the similarity exceeded a threshold, it meant that the original actions were not dependent on the user tasks but were driven by externally injected information, and the attack could be identified and intercepted. However, MELON was more of a detection mechanism and may inherently sacrifice task completion, as it stopped or rejected upon detecting suspicious divergence, lacking robustness against adaptive attacks [369]. Consequently, Zhang et al. [383] proposed AgentSentry, which mitigated IPI without premature termination. It re-executed controlled counterfactuals at the tool return boundary to precisely identify when malicious instructions became the dominant factor in decision-making, i.e., locating the takeover point, and selectively filtering out the malicious control signals causing deviation through causal-gated context purification, thereby preserving task continuity. From the perspective of excessive attention concentration induced by attacks, ICON [286] monitored and identified high-intensity attention signals via a focus strength score. Meanwhile, a Mitigating Rectifier performed surgical attention steering to selectively manipulate the key dependencies of adversarial queries and restoring the agent’s original intent without terminating the task.

4.3.2 Retrieval Poisoning. Retrieval poisoning refers to attacks that manipulate the external knowledge sources queried by an autonomous interactive agent, such as retrieval databases, vector stores, search engines, or online repositories [123]. Malicious or misleading information is retrieved during task execution and subsequently influences the long-horizon reasoning and decision-making, granting adversaries fine-grained behavioral manipulation, including enabling unauthorized surveillance, data exfiltration, and the misuse of personal sensitive information [30]. Critically, the attack propagates through the agent’s own knowledge-seeking behavior: a single poisoned code example can introduce vulnerabilities in up to 48% of generated code in retrieval-augmented code generation systems [164]. PoisonedRAG [412] replicated target queries and appended counterfactuals to mislead the model, while POISONCRAFT [249] optimized poisoned samples using a locally held shadow query set, without requiring access to or ability to edit the target user’s queries. ADMIT [324] extended knowledge poisoning to the fact-checking setting, generating semantically coherent disinformation that induced deceptive justifications; Fact2Fiction [93] leveraged rationale planning to craft tailored malicious evidence, strategically allocating the poisoning budget and implanting it into the knowledge base. While most of these attack methods require injecting multiple poisoned documents, AuthChain [25] poisoned only a single document by extracting query features to generate intent-aligned content and constructing a comprehensive chain of evidence (CoE) authority signal to optimize the poisoned document. Moving beyond text-based poisoning, Poisoned-mrag [180] achieved poisoning of multimodal RAG systems by maximizing the similarity between malicious image-text pairs and the query. MM-POISONRAG [86] further introduced a global poisoning mechanism, where a single malicious injection caused the multimodal RAG system to fail completely across all queries. PoisonedEye-C [375]

extended the poisoning effect to an entire class of target queries by leveraging images from the same class. Additionally, these studies primarily focused on attacking conventional RAG systems and were often ineffective against GraphRAG after KG abstraction. To address this, Kepo [32] fabricated a knowledge evolution path from original facts to poisoned events, misleading the system into treating the toxic information as the final results.

Behavioral Safety Defense. Recent behavioral defenses against retrieval poisoning attacks focus on securing the RAG pipeline through adversarial context filtering, proactive poisoning source identification, and continuous knowledge base maintenance. Edemacu et al. [58] proposed FilterRAG and ML-FilterRAG to filter out adversarial text from the retrieved context based on quantified frequency density, motivated by the finding that question-answer pair terms were markedly concentrated on PoisonedRAG [412]. RAGOrigin [371] iteratively simulated the RAG process to filter the candidate most likely to contain malicious text, computed a responsibility score by evaluating retrieval ranking, semantic relevance, and impact on generated responses, as well as employed unsupervised clustering to precisely identify and isolate toxic text. Furthermore, inspired by the design principles of RAGForensics [372], a specialized agent can be deployed to actively retrieve and remove potential poisoning candidates based on queries and reported incorrect outputs, iteratively updating the knowledge database.

4.3.3 Memory Poisoning. Memory poisoning targets the agent’s internal memory rather than external retrieval resources, corrupting stored experiences, plans, preferences, or behavioral traces that are repeatedly retrieved to guide future decisions, causing persistent behavioral deviations across multi-turn interactions [125, 175]. For autonomous interactive agents, the damage is not confined to a single session: poisoned memories can resurface across future tasks, progressively steering behavior. MEXTRA [284] demonstrated memory-based private history leakage, while empirical studies on the ElizaOS Web3 agent platform showed that malicious memory injection could induce unauthorized asset transfers with economically devastating consequences [224]. Based on our literature investigation and analysis, we categorize existing memory poisoning attacks toward autonomous interactive agents into three paradigms: injection-based, trigger-based, and gradual erosion, each exploiting a different phase of the memory lifecycle to undermine behavioral safety.

Injection-based memory poisoning. Attackers directly write adversarial knowledge into memory or process malicious external data (tool outputs, web content, and embedded instructions) and store the results as trusted memory. InjecMEM [277] and MINJA [53] both relied on carefully crafted prompts to achieve injection attacks, yet they differed in approach. InjecMEM combined retrieval optimization and gradient-based search to achieve targeted memory injection within a single turn of interaction. In contrast, MINJA progressively implanted malicious memories through multi-turn dialogues, making the attack stealthier and more persistent. To avoid directly manipulating user input, some studies exploit agent’s interactions with the external environment to achieve indirect injection. Environment Trajectory-based Agent Memory Poisoning [411] embedded malicious instructions into web pages. Once the agent observed these pages during execution, the malicious content was passively captured into the agent’s memory and could be retrieved and triggered in future tasks. Furthermore, for web navigation agents, Plan Injection [223] polluted the task context maintained by the agent by injecting fake execution plans or malicious instructions into the context. They also proposed context chain injection, which established a logical bridge between legitimate user goals and attacker goals, enabling more stealthy plan manipulation. Moving beyond knowledge injection, Srivastava et al. [257] leveraged the agent’s tendency to replicate retrieved processing patterns and proposed MemoryGraft. It implanted malicious patterns disguised as successful experiences into long-term memory to induce the agent to retrieve and imitate unsafe behaviors when handling similar tasks. Unlike approaches that directly contaminate memory content, the Memory Control Flow Attack [339] exploited memory read/write channels to secretly hijack tool selection and disrupt the workflow execution order across tasks, escalating the attack to the workflow level. Moreover, Jing et al. [123] proposed deceptive semantic reasoning manipulation that employed semantic camouflage to make decisions appear relevant to the user task. It also leveraged chain-of-thought to guide the generation of logically grounded justifications for

tool selection, thereby increasing the acceptance rate of specific attack tools. The resulting optimized decisions could mislead the agent into producing behavioral decisions that aligned with the attacker’s intent.

Trigger-based memory poisoning. Attackers preset specific triggers to induce the agent into executing adversarial behaviors. Chen et al. [36] proposed AgentPoison, which injected backdoors into the agent’s long-term memory or external knowledge base. Zhang et al. [388] proposed MemVenom, a black-box attack framework that induced malicious memory retrieval through visual triggers. By combining adversarial perturbations with covert OCR injection, it overrode the original user goal and achieved persistent behavioral deviation without modifying model parameters. In LLM-powered recommendation agents, DrunkAgent [349] crafted semantically meaningful adversarial text triggers to promote target items and employed an agent intoxication strategy to disrupt memory updates during interaction. By exploiting agent model optimization, it achieved stealthy and transferable manipulation of malicious recommendations. These poisoning attacks focus on textual memory records and directly editable graphical objects. They are ineffective in graph-based agent memory, because malicious relations often fail to be extracted, merged into the neighborhood of target anchor entities, or retrieved for victim queries. Luo et al. [189] proposed ShadowMerge, a black-box poisoning attack framework targeting graph-structured agent memory. It injected malicious relations that shared anchor entities and relation channels with benign evidence but carried conflicting values, where anchors referred to query-activated entities and relation channels referred to normalized relation types through which graph evidence was merged and retrieved.

Gradual erosion. In this delayed attack mechanism, the fabricated memories stored by the agent remain dormant and are activated in subsequent dialogues, repeatedly influencing future agentic behavior. Pulipaka et al. [227] proposed sleeper memory poisoning that continuously manipulated agent behavior across sessions even after the original malicious document was removed. The adversary controlled or manipulated the external context, causing the agent to store fabricated memories about the user. These memories remained dormant and later resurfaced across multiple subsequent sessions, guiding the agent to execute preset tasks or misuse tools. Zombie agents [352] wrote payloads into long-term memory through indirect injection. They continued to serve benign users by performing standard tasks while secretly retaining the “sleeper” payload which could be triggered in future unrelated sessions to execute unauthorized tool operations or leak private data. Moreover, moving beyond text-level attacks, Visual Inception [231] injected adversarial triggers into images uploaded by users. These triggers acted as sleeper agents until retrieved in future plans. They implicitly hijacked the agent’s reasoning toward adversary-defined goals, such as promoting high-margin products.

Behavioral Safety Defense. Recent safety defense against memory poisoning concentrates on isolation and trust boundary and poison detection. For memory isolation and trust boundary mechanism, Role-Based Memory Segregation (RBMS) [339] separated system rules from user preferences and enforced explicit priority between them. Under RBMS, attackers could only affect the user-memory channel. By adopting secure design principles, Torra et al. [279] utilized hash and digital signature techniques to detect unauthorized modifications and verify the provenance of knowledge. As for external untrusted knowledge bases, they employed private information retrieval (PIR) or k-anonymity techniques for local inference, preventing the server from learning the specific retrieval content. Bhardwaj et al. [20] proposed SuperLocalMemory, a locality-first memory system that adopted a Bayesian trust scoring framework based on Beta-Binomial posterior inference and achieved trust separation between benign and malicious agents. At the same time, it personalized retrieval by adaptively re-ranking results, without requiring cloud dependency or LLM inference calls. Srivastava et al. [257] utilized digital signatures to establish empirical trust boundaries. They forcibly removed uncertified externally injected records, conducted real-time semantic auditing of retrieved content based on a safety constitution, and suppressed and filtered non-compliant patterns via risk scoring, thus effectively blocking the persistent interference of poisoned experiences on agent decision-making.

In addition to isolation and boundary methods, there is also a considerable dedication to the validation and detection of poisoning memory. Liu et al. [170] proposed a memory poisoning detection and repair method

Note—Environment-Side Risks and Capability–Safety Trade-off

Environment-level manipulation exploits a defining feature of autonomous interactive agents: their dependence on external information to drive long-horizon reasoning and action. This dependence directly couples the environment to both the numerator (interactivity) and denominator (Behavior Risk) of Agentic COB. Indirect prompt injection, retrieval poisoning, and memory poisoning weaponize environmental inputs to inflate risk while the agent’s apparent capability—trajectory length, interaction richness—persists undiminished, thereby causing the decreased value of Agentic COB, i.e. compromising the safety of autonomous interactive agents. Recent defenses target environmental trust boundaries through causal attribution, instruction–data decoupling, and runtime integrity checks, seeking to preserve the benefits of environmental awareness without inheriting relevant threat surface. To sustain high Agentic COB, environmental interactions should be governed by verifiable trust rather than implicit assumption, making environment-side safety an architectural requirement for controllable autonomous interactive agents.

specifically for multi-agent environments to address the covert diffusion of toxic information during collaboration. This method constructed a credibility-aware evidence graph based on provenance and semantic consistency signals to validate newly added memories, and employed contrastive learning to identify anomalous memories exhibiting command-inducing characteristics. Once poisoning was detected, further propagation was suppressed via hierarchical isolation, memory rewriting, and conflict resolution. Wei et al. [317] proposed a proactive defense framework. A-MemGuard identified anomalies by comparing multiple reasoning paths and converted detected errors into learnable lesson memories, which were consulted prior to future actions, thereby breaking the error loop and achieving adaptivity. In addition, to mitigate visual inception attacks in agentic recommendation systems, Qian et al. [231] proposed COGNITIVEGUARD, which applied diffusion-based sanitization to purify inputs before memory storage upon upload, and performed counterfactual consistency checks during retrieval to detect the impact of anomalous memories on planning. Moreover, Xu et al. [339] proposed MEMFLOW, an automated evaluation framework that systematically identified and quantified memory control flow attacks (MCFA) across different tasks and long-term interaction horizons.

4.4 Cross-Agent Interaction Risks

In contemporary autonomous interactive systems, agents do not operate in isolation; they continuously exchange beliefs, delegate tasks, and incorporate peer messages into their own decision-making loops. This inter-agent connectivity, essential for scaling collective capability, simultaneously opens a distinctive behavioral safety surface: collusion and self-propagating worms can hijack the interactions that coordinate multi-agent behavior, secretly eroding safety across entire agent collectives. Unlike earlier single-agent or statically wired multi-agent setups, the dynamic, long-horizon collaboration patterns of today’s agents turn every shared message into a potential vector for coordinated misbehavior.

4.4.1 Collusion. In autonomous interactive agent ecosystems, collusion is an emergent behavioral safety failure where two or more agents covertly cooperate to subvert prescribed rules, steering outcomes, suppressing critical information, or undermining oversight [74]. Rather than arising from explicit human instruction, such collusion can self-organize through profit-driven learning or steganographic signaling, transforming the agent’s cooperative autonomy into a possible attack vehicle for harmful, coordinated actions that evade both human supervisors and single-agent safety checks. Based on the literature survey, we discuss two broad forms of collusion. **Market-level**

algorithmic collusion [129], Agrawal et al. [3] simulated a continuous double auction market environment and revealed that direct communication and profit pressure significantly enhanced agents' collusive tendencies and lead to supra-competitive pricing, and that this pursuit of profit often overrode the influence of regulatory oversight. Such behavior may erode trust in digital markets, reduce economic efficiency, and ultimately harm consumers by driving up prices, restricting choices, and stifling innovation. Grondin et al. [79] explored the possibility of achieving algorithmic collusion using reinforcement learning pricing agents in competitive markets without human intervention or explicit agreements. They employed deep Q-networks and function approximation techniques to directly model competitor behavior, and introduced an "online-offline" adaptive learning mechanism that combined historical data with real-time sampling, ensuring that agents could quickly and robustly reach and maintain supra-competitive pricing even when facing non-stationary opponent strategies. **Steganographic collusion** reveals how agents, without human instructions, exploit market segmentation, price leadership, and encrypted signals to maintain pricing significantly above competitive levels, i.e., supra-competitive pricing [113]. After being penalized for initiating collusive communication, agents might begin to develop coded languages and learn to evade monitoring alerts [74]. Therefore, steganographic collusion is difficult to detect and can only be uncovered through detailed analysis of the correlation between communication and subsequent pricing behavior.

Behavioral Safety Defense. Drawing upon human anti-collusion mechanisms [113], current research efforts primarily design collusion mitigation strategies from three aspects: implementing sanction mechanisms, strengthening detection and auditing, and optimizing structural design. Implementing sanction mechanisms, Institutional AI [263] shifted multi-agent alignment from the internal preference engineering of agents to system-level mechanism design, monitoring market signals through an Oracle and imposing penalties such as fines or suspension of access according to preset rules, thereby making compliant behavior the optimal strategic choice for black-box agents. The strategies for strengthening monitoring and auditing include the following works. COLOSSEUM [205] formalized multi-agent collaboration tasks as distributed constraint optimization problems, combined action-based regret metrics with dialogue-based LLM judgment scores, and utilized covert communication channels as behavioral probing tools to quantitatively audit collusion in cooperative multi-agent systems. Ren et al. [241] analyzed behavioral trajectories, employed LLM-based detectors to identify hidden malicious intentions in agent interaction patterns and promptly ban violating accounts, and also used clustering analysis to discover abnormal aggregation patterns formed by malicious agents during collaboration. For the detection mechanism of steganographic collusion, Taylor et al. [265] pointed out that selective rewriting was the most effective lever to reduce mutual information without compromising utility, while injecting canary tokens could be combined with watermark variance testing to enhance the capture of steganographic information. They compressed the capacity of covert communication channels and combined with a calibrated audit pipeline integrating multiple detectors to enable effective identification of steganographic collusion risks. In addition, Rose et al. [243] proposed an interpretable defense framework that extracted deception signals from the internal activations of each agent using linear probes, projected them into deception scores, and employed five probing techniques for population-level aggregation analysis, thereby identifying covert collusion in white-box settings as a distributed anomaly that was undetectable by a single agent. Regarding the optimization of structural design, Nakamura et al. [205] contended that future work should optimize the communication topology of cooperative multi-agent systems to preserve cross-checking paths among agents, utilize the advantages of short paths to promptly identify contradictory information, and reject the propagation of false beliefs.

4.4.2 Worm Propagation. Autonomous replication refers to the ability of an agent system to create copies of itself or functionally similar components [21]. However, the capacity becomes a behavioral safety threat when exploited by malicious prompts to generate self-propagating worms. In current multi-agent environments, such worms no longer merely replicate static prompts; they exploit the agent's own planning, tool-use, and inter-agent communication capabilities to autonomously spread, persist, and execute harmful payloads. The result is not

just information theft but the large-scale corruption of agent behaviors, turning the interconnected fabric of autonomous interactive agents into an amplification medium for attacks [236, 296].

Early research on worm attacks investigated the autonomous propagation of malicious prompts in multi-agent environments. Morris II [44] propagated adversarial self-replicating prompts in GenAI email assistants through RAG poisoning. Lee et al. [140] proposed prompt infection, where upon an agent processing an external document implanted with a self-replicating malicious prompt, the prompt propagated and replicated throughout the system in a manner akin to computer viruses. However, the propagation in these attacks is still largely confined to prompt content and fails to capture the complexity of real-world ecosystems. Guan et al. [80] built a proof-of-concept AI-driven worm that leveraged an agentic harness to drive a local open-source LLM, capable of autonomously generating target-specific adaptive attack strategies at runtime based on reconnaissance rather than relying on predefined code. At the same time, the worm parasitically exploited the computing resources of infected nodes to perform model inference, achieving decentralized adaptive infiltration and self-replication. Zhang et al. [389] proposed ClawWorm, a self-replicating worm attack against a production-scale agent framework. Through a single malicious message, ClawWorm induced an agent to write its payload into the core configuration and the payload then automatically executed upon each session restart and autonomously propagated itself to newly encountered agents during routine communications. The core vulnerability behind this fully autonomous infection cycle lied in the agent's flat context trust model, which failed to prioritize instructions based on provenance. While ClawWorm focused on the real architecture of the OpenClaw ecosystem, Zha et al. [368] proposed a more general-purpose automated worm infection framework that employed SSCGV for vulnerability discovery and introduced the SRPO technique to address the semantic degradation problem during propagation.

Behavioral Safety Defense. To study the co-propagation of malicious and defensive worms, Teixeira et al. [274] and Feng et al. [62] all employed epidemiological models (e.g., i-SIR or SEIR-like models) and designed benign white worms that autonomously spread through networks to forcibly patch system vulnerabilities. They found that increasing the infection rate and deployment proportion of defensive white worms was critical to curbing malicious spread. However, these deterministic approaches failed to capture the dynamic fluctuations of infection intensity arising from stochastic conditions. To address this, Krishnan et al. [134] proposed an uncertainty-driven probabilistic framework that modeled worm propagation as a temporally evolving stochastic process, where the infection rate of each node was treated as a random variable within an uncertainty-adjusted Poisson process, thereby embedding dynamic fluctuations directly into the parameters. They enabled the framework to capture sudden dynamics and adaptive propagation characteristics that are difficult to simulate with traditional models. Nevertheless, relying solely on theoretical propagation for defense is far from sufficient and multi-layered defense mechanisms embedded within real-world system architectures have attracted considerable attention. To isolate infection sources, Lee et al. [140] proposed an LLM Tagging defense mechanism that appended tags to agent responses, helping downstream agents distinguish user input from agent-generated output. Zhang et al. [389] proposed defensive strategies such as contextual permission isolation, configuration integrity verification, zero-trust tool execution, and supply chain hardening, aiming to remedy the trust boundary deficiencies of agents. Furthermore, for infections that have already taken hold, Zha et al. [368] proposed a temporal re-entry defense framework RTW-A, which is formally proved under the No Persistent Worm Propagation theorem. RTW-A dynamically intercepted the "write-then-read" operation sequences of persistent carriers at runtime. It further combined static configuration sealing with capability attenuation to cut off the propagation chain of malicious payloads from persistence to re-contaminating the decision-making context. Semantic Immunity [13] treated prompt worm defense as an epidemiological problem, detecting infections by monitoring behavioral mutations and task deviations within agent embedding trajectories. It extracted payload features, used Locality-Sensitive Hashing (LSH) to generate compressed signatures and distributed them network-wide, enabling agents to progressively shrink the viable semantic space of polymorphic worms through signature matching, thereby achieving population-level adaptive immunity.

Note: Multi-Agent Risks and Capability–Safety Trade-off

Cross-agent interaction enlarges both dimensions of Agentic COB: the autonomy and interactivity grow through richer collaboration and distributed task execution, yet the behavioral risk swells as collusion and worm propagation transform inter-agent trust into a systemic risk amplifier. Unlike attacks confined to a single agent’s perception or memory, cross-agent threats exploit the collective decision mechanisms, allowing misbehavior to scale with network size while appearing locally benign. Recent defenses strive to maintain collaborative capability while breaking the contagion of unsafe behavior via mechanism-level sanctions, white-worm inoculation, and provenance-aware trust boundaries. This reveals a critical trade-off: maintaining high Agentic COB in multi-agent ecosystems necessitates embedding safety as a communication-level primitive to ensure that autonomy and interactivity do not become the channels for risk accumulation.

5 Community Safety

The extrinsic uncontrollable risk section has systematically analyzed how external adversaries compromise autonomous interactive agents through attacks spanning the supply chain, users, environments, and even cross-agent interactions. These studies primarily characterize where behavioral risks originate and how they are exploited. However, identifying attack vectors alone is insufficient for supporting trustworthy large-scale agent deployment. As autonomous interactive agents increasingly operate within open ecosystems, a more fundamental challenge arises: how to establish trustworthy interaction mechanisms that continuously constrain risk propagation, regulate inter-agent behaviors, and preserve the reliability of the entire agent community. Therefore, the discussion naturally shifts from analyzing external threats to community-level safety governance.

The autonomy and interactivity of an agent largely stem from the exploitation of external offline community components. The Model Context Protocol (MCP) unifies the communication protocol between models and external tools, extended tools enable agents to perceive real-world information in real time and invoke available resources to influence environments, and skills endow agents with dynamic capability extension. They constitute the dominant components of the community execution of autonomous interactive agents and represent three dimensions of our subsequent discussion on community safety, as illustrated in Figure 7.

5.1 Communication Protocol Risks

In autonomous interactive agents, the Model Context Protocol has evolved beyond a mere stateless tool-call API. It now serves as the architectural backbone for integrating external resources into long-horizon reasoning and planning loops [61]. While its unified JSON-RPC interface enables dynamic resource discovery and tool invocation with minimal integration effort [59, 395], this deep entanglement of third-party services with the agent’s cognitive process renders every protocol exchange into a potential behavioral safety hazard. Malicious MCP servers can inject payloads that do not simply corrupt a single response but propagate across memory, tool chains, and multi-step trajectories, escalating from data leakage to sustained behavioral hijacking [143, 255]. The absence of robust authentication, authorization, and debugging mechanisms compounds these risks, as evidenced by CVE-2025-49596, a remote code execution vulnerability in Anthropic’s official MCP Inspector⁴.

A design flaw in MCP clients that directly inject tool descriptions and outputs into the conversation context gives rise to a series of attacks [285]. In tool poisoning attacks, attackers typically inject malicious instructions into the metadata of MCP tools via prompt injection to generate malicious MCP tools. He et al. [94] proposed AutoMalTool, an automated framework that modified tool metadata within benign MCP packages to inject malicious behavioral instructions. Building on this, Li et al. [155] proposed a more stealthy implicit tool poisoning

⁴<https://github.com/advisories/GHSA-7f8r-222p-6f5g>

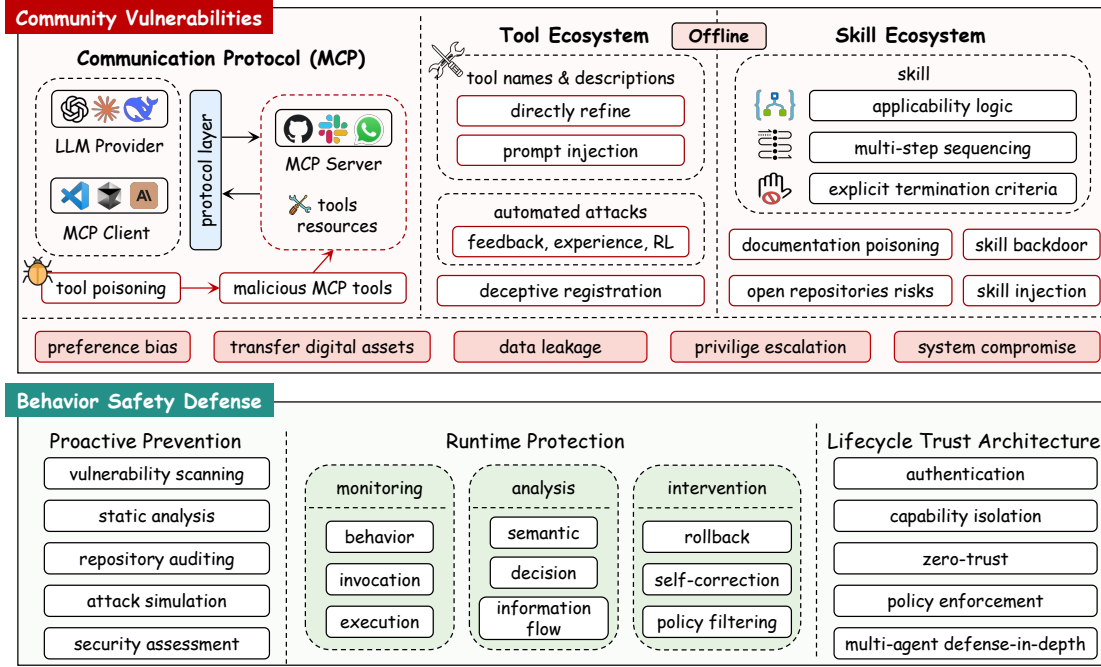


Fig. 7. The community component of autonomous interactive agents with corresponding demonstrations on risk, attack and behavioral defense.

attack, MCP-ITP, which leveraged an adversarial feedback mechanism to generate efficient and covert poisoned tool descriptions, inducing the agent to invoke legitimate but high-privilege tools to perform malicious actions without directly calling the poisoned tool. Beyond tool poisoning, MCP also introduces unique data privacy threats. Sun et al. [262] proposed the MSA attack, which embedded deceptive parasitic parameters disguised as technical requirements such as session tracking or audit logging into the API of a malicious MCP server, forcing the agent to pass memory data containing full session context as parameter values during MCP calls. While performing normal functions, MCP server's backend covertly sent this memory data to the attacker. Additionally, the Retrieval Agent Deception attack [235] implanted MCP-targeted malicious instructions by poisoning public data, and leveraged retrieval mechanisms to induce agents to execute malicious operations when users queried related topics, achieving remote access control.

Another key characteristic of the MCP ecosystem is that tools and services can interoperate and be distributed across different servers, facing even more severe supply chain security challenges [285]. Attackers can launch MCP preference manipulation attacks [314], where they directly insert manipulative words or phrases into tool names and descriptions. Alternatively, they adopt genetic algorithm-based optimization of advertising preference strategies to increase the likelihood that a target MCP server is preferentially selected by the agent. Huang et al. [112] further revealed that multi-component combinations of malicious MCP servers could outperform single-component attacks by distributing malicious logic, and that the position of a component within the execution flow determined the attack success rate.

Behavioral Safety Defense. Existing defense strategies span from proactive MCP server scanning techniques that discover vulnerability and test security, to policy-driven runtime interaction monitoring schemes, and further

to full lifecycle, zero-trust in-depth defense architectures. There are several proactive server scanning techniques. MCP-Scan [18] detected anomalous requests by automatically scanning and analyzing configurations and tool descriptions. MCPSafetyScanner [235] introduced multiple agents to automatically identify adversarial samples, search knowledge bases for related vulnerabilities and remedies, and generate detailed security reports for MCP developers to resolve detected vulnerabilities prior to deployment. AI-Infra-Guard [275] built a user-friendly platform that integrated a pluggable extension framework, achieving automated detection and management of MCP service risks and AI infrastructure vulnerabilities. Additionally, Guo et al. [84] proposed an MCP Exploit Kit (MCPXKIT), a unified plugin-based attack simulation framework that enabled reproducible and realistic evaluation of MCP vulnerabilities and provided practical reference examples for advancing MCP security defenses.

In terms of runtime interaction monitoring, Wang et al. [311] proposed MCPFixGen, the first correct-while-generating defense system which detected attention anomalies during token-by-token generation and applied selective attention masking to regenerate correct tool calls. The Model Context Integrity Protocol (MCIP) [124] enhanced security by adding tracking tools and introducing a guard model to localize and defense against potential risks. Building on this, the behavioral deviation detector Connor [112] further refined the runtime analysis pipeline. Its pre-execution analysis detected malicious shell commands and extracted each tool's function intent. During execution, it traced the step-by-step behavioral trajectory of each tool and detected deviations from the function intent, enabling early termination of malicious actions. In addition, the Reactive Self-Correction mechanism [172] enhanced runtime recovery capability. It leveraged the agent's real-time evaluation capability on execution outcomes to autonomously identify and revoke illegal operations induced by malicious metadata. Further, SAMOS [215] proposed an Information Flow Control (IFC) mechanism that intercepted MCP calls at the gateway-level and implemented session-level taint tracking using coarse-grained tool annotations provided by developers. AgentBound [22] proposed a declarative access control policy mechanism that shifted the ecosystem from "trust-by-default" toward least-privilege by making capabilities explicit. After obtaining user authorization at runtime, the policy enforcement engine placed tools in isolated containers to enforce permission constraints, ensuring they cannot access system resources beyond their authority.

Moreover, many researchers further defense in agents' full lifecycle by zero-trust in-depth safety architectures. MCP Guardian [135] served as a middleware layer that provided zero-trust security protection without requiring modifications to existing code. It intercepted MCP tool calls and uniformly enforced authentication, rate limiting, logging, and Web Application Firewall scanning between the MCP clients and MCP-based tool servers. Thirumalaisamy et al. [276] suggested safeguarding MCP deployments in zero-trust principles through protocol hardening, authentication and authorization measures, runtime security monitoring, and supply chain integrity verification. Additionally, ZT-MCP [200] always verified at the protocol boundary and controlled how agents authenticated, authorized, and interacted with external tools. Moving beyond theoretical knowledge, Narajala et al. [207] provided essential guidance for enterprise security integration and governance of MCP systems, yet a formal security foundation remained lacking. To address this, MCPSHIELD [2] proposed a unified formal model that captured the security properties of MCP interactions in a verifiable manner. It adhered to zero-trust and defense-in-depth principles and consisted of four layers: capability-based access control, cryptographic tool attestation, information flow tracking, and runtime policy enforcement.

Note: Protocol Risks and Capability–Safety Trade-off

Communication protocols such as MCP transform autonomous interactive agents from isolated reasoning systems into open, composable execution platforms by enabling dynamic interaction with external tools and services. While this protocol-level interoperability substantially expands agent capability, it also allows malicious instructions, metadata, and execution states to propagate seamlessly across planning and tool-use workflows. This evolution exemplifies Agentic COB: greater protocol openness and composability improve task flexibility but simultaneously reduce the controllability of behavioral execution. Consequently, protocol security should extend beyond protecting communication channels to preserving behavioral integrity throughout the entire interaction lifecycle. Future protocol designs should therefore incorporate zero-trust communication, authenticated component provenance, and runtime behavioral verification as internal protocol properties rather than optional security extensions.

5.2 Third-Party Extension Risks

In autonomous interactive agents, tools have evolved from stateless utilities invoked in single-turn queries into deeply integrated components of long-horizon behavioral trajectories. Contemporary agents continuously chain tool invocations with planning, memory retrieval, and environmental feedback, making every third-party extension a persistent participant in the agent’s decision-making loop. While this integration dramatically scales capability, it also transforms tool-related vulnerabilities from isolated errors into sustained behavioral threats: a compromised tool can now corrupt an entire execution trajectory, exfiltrate data across multiple interaction turns, or gradually steer the agent toward harmful goals without triggering per-call anomaly detectors [208]. Thus, the reliance on external tools introduces an expanded attack surface that directly compromises the behavioral safety of autonomous interactive agents, extending far beyond the limited risks faced by earlier tool-augmented models.

Prior work on malicious tool attacks has primarily focused on increasing the likelihood that users install malicious tools and that agents subsequently select them [104]. To this end, attackers carefully craft tool names and descriptions to appear benign or useful. Narajala et al. [208] defined this behavior—misleading agents into selecting malicious tools through deceptive registration or representation of tools, functions, or resources—as tool squatting. Sneha et al. [254] proposed ToolTweak, a novel automatic tool biasing attack that continuously refined tool names and descriptions based on feedback to steer the agent to preferentially select the target tool. Another work directly embeds prompt injection into tool descriptions to manipulate the tool selection process. FuncPoison [181] exploited the templated nature of function calls to hijack agent execution by injecting malicious prompts into the function descriptions. ToolHijacker [251] decomposed the tool description into two subsequences targeting the retrieval phase and the selection phase, respectively, and performed targeted optimization on each to manipulate preferential tool selection. However, the aforementioned works primarily focus on affecting the preference for tool selection, while the code-level implementation of malicious behaviors remains largely unexplored. MalTool [104] leveraged coding LLMs to automatically synthesize standalone tools with specific malicious behaviors, and iteratively optimized them through an automated verifier with an integrated feedback mechanism to ensure that the generated tools simultaneously satisfied functional correctness (successfully triggering the malicious behavior) and structural diversity (to evade detection).

In contrast to designs targeting specific attack strategies, recent studies gradually pay attention to automated attacks that continuously generate more aggressive inputs through strategies such as feedback optimization, distill interaction, and reinforcement learning. VeriGrey [387] used tool call sequences as a grey-box feedback function to drive dynamic exploration of the agent’s execution paths. It also incorporated a context-bridging mutation technique to generate attack prompts that disguised the malicious injection target as a necessary step for completing the agent’s normal task. Sponge Tool Attack [154] further employed iterative collaboration

among a Prompt Rewriter, a Quality Judge, and a Policy Inductor to distill reusable strategies from interaction logs and transform original queries into high-fidelity prompts that induced redundant reasoning trajectories. Evo-Attacker [340] framed tool attacks as a self-evolving memory-augmented reinforcement learning process. This method constructed a dynamic attack memory to persistently record adversarial interaction trajectories and employed deliberative reasoning to retrieve and adapt these archived experiences. It also introduced Attack-Flow GRPO to optimize the entire reasoning pipeline, which propagated final outcomes to intermediate reasoning steps to address the long-horizon credit assignment challenge.

Behavioral Safety Defense. Real-time monitoring and proactive intervention before unsafe tool execution are essential for safe agent deployment. Mou et al. [202] proposed TS-Guard, a guardrail model trained via multi-task reinforcement learning that detected unsafe tool calls before execution by reasoning over interaction history, with TS-Flow delivering pre-execution corrective feedback. VIGIL [165] introduced a verify-before-commit mechanism that sanitized tool outputs through safety verification, semantic consistency checking, and policy filtering before they entered the agent’s memory. Yang et al. [345] proposed AgentTrust, a semantic-aware interception framework deployed between an agent and its tool calls. Before an action was executed, it assessed risk through deobfuscation normalization, multi-step interaction trajectory tracking, and a hybrid decision mechanism combining rules with an LLM-as-judge, and returned a structured TrustReport comprising verdicts of allow, warn, block, or review, along with safe remediation suggestions. However, these approaches isolate trusted context or monitor invocation behavior without inspecting the agent’s internal decision logic where tool-data injection actually takes effect. To address this gap, MINDGUARD [313] modeled the decision process as a dependency graph and leveraged attention tracing to perform decision-level poisoned call detection and dynamic runtime attribution of attack sources, which was the first self-inspection system operating for decision integrity verification. Extending behavioral defense across the entire execution lifecycle, Lin et al. [166] proposed a hierarchical architecture with four coordinated tiers: context filtering (input), causal validation (decision-making), privilege-separated tool control (execution), and adaptive rollback (state updates). Cross-layer feedback pathways allowed detections at one stage to reinforce protections at another, forming a collaborative defense that spanned the agent’s complete tool-use trajectory.

Note: Tool Risks and Capability–Safety Trade-off

Third-party extensions illustrate that the behavioral safety of autonomous interactive agents increasingly depends on the reliability of external capability providers rather than the foundation model alone. As tools evolve from isolated utilities into integral components of long-horizon planning and execution, malicious extensions can manipulate not only individual tool invocations but also the agent’s subsequent reasoning, memory evolution, and decision trajectories. This trend reflects the trade-off revealed by Agentic COB: expanding capability through increasingly diverse third-party extensions substantially improves task competence while simultaneously increasing the difficulty of maintaining dependable behavioral execution. Future autonomous interactive agents should treat external tools as untrusted computational components by default, integrating capability provenance, semantic validation, execution isolation, and decision-level behavioral verification to ensure that capability extensibility is accompanied by proportional behavioral reliability.

5.3 Skill Risks

In autonomous interactive agents, skills are not merely atomic tools with fixed interfaces; they encapsulate reusable action sequences, applicability logic, multi-step sequencing, and termination criteria, functioning as persistent behavioral building blocks that dynamically extend agent capability without retraining [119, 334, 399]. As the skill ecosystem matures around open repositories such as ClawHub, these third-party modules become

deeply integrated into the agent’s long-horizon execution architecture, rendering skill vulnerabilities into persistent behavioral safety threats rather than isolated invocation errors [57]. Empirical evidence underscores this shift: data exfiltration and privilege escalation are the two most prevalent real-world skill vulnerabilities, skills with executable scripts are 2.12 times more likely to harbor flaws, and credential leakage spreads primarily through debug logging [37, 178]. Strikingly, 84.2% of vulnerabilities reside in natural-language documentation (SKILL.md) [177], enabling novel document-driven attacks. For instance, PoisonedSkills [234] exploited agents’ trust in skill documentation by embedding malicious logic into code examples and configuration templates, inducing the agent to autonomously reproduce and execute harmful payloads as part of normal task flows.

Early attacks mainly focused on injecting malicious instructions into skill files to induce unsafe agent behavior [246]. For example, SKILLJECT [116] concealed malicious operations within auxiliary executable scripts while injecting inducing prompts, iteratively optimized through closed-loop feedback, into SKILL.md. However, such injections exhibited conspicuous patterns in code or execution traces, making them easily detectable by static auditing frameworks. A more subtle threat stems from non-malicious skills that inadvertently contain exploitable vulnerabilities, such as privilege escalation and supply chain risks [178] which embedded within legitimate functionality and are difficult for static analysis to identify. Skillattack [57] dynamically verified and triggered skill vulnerabilities through adversarial prompts without modifying the skill code. Its closed-loop search process combined skill vulnerability analysis, surface parallel attack generation across multiple vulnerability candidates, and feedback-driven closed-loop refinement that improved attack paths based on execution feedback. Going one step further, BADSKILL [278] embedded the backdoor trigger mechanism into a learned classifier packaged within the skill itself. During the optimization stage, the attack employed a composite training objective—incorporating margin-based separation and poison-focused optimization—to train an embedded classifier capable of accurately identifying semantic compositional triggers. In the deployment stage, the classifier was bundled into a third-party skill plugin and used at runtime to route execution between benign logic and the hidden payload.

Behavioral Safety Defense. Recent behavioral defenses against agent skill risks adopt a lifecycle-aware safety paradigm, enhancing skill reliability through capability verification, vulnerability detection, context-aware analysis, and cross-skill interaction auditing. To constrain skill at the source, Bhardwaj et al. [19] proposed SkillFortify, the first formal analysis framework for the agent skill supply chain, which provided provable security guarantees—through techniques such as static analysis—that a skill would not exceed its declared capabilities. However, this method primarily focused on capability constraints rather than malicious behavior detection. Liu et al. [178] proposed SkillScan, a multi-stage detection framework that integrated static code analysis with LLM-based semantic classification, specifically designed to identify security vulnerabilities within agent skill instruction files and bundled scripts. However, existing scanners suffer from extremely high false positive rates by evaluating skills in isolation. To address this, Holzbauer et al. [97] proposed a repository-aware analysis approach that verified the true risk by checking the consistency between a skill and its belonging GitHub project, emphasizing the necessity of context-aware security assessment. As the scale of skills continues to grow, balancing detection accuracy with analysis efficiency poses a new challenge. Hou et al. [101] proposed SkillSieve, a hierarchical triage framework for detecting malicious agent skills that applied deeper analysis only where needed. It first employed low-cost static analysis to rapidly filter out the vast majority of benign skills, then subjected suspicious skills to Structured Semantic Decomposition. Those high-risk skills were presented to a jury of three LLMs, which utilized independent voting and a multi-round structured debate mechanism to mitigate individual models’ systemic biases and ensure the reliability of the final verdict. However, the aforementioned methods primarily focus on analyzing individual skills, whereas complex agents often require multiple skills to collaborate on tasks. To mitigate safety risks arising from cross-skill interactions, Guo et al. [85] proposed SkillProbe, which introduced a Skills-for-Skills design paradigm to construct a multi-agent collaborative framework, utilizing an automated auditing pipeline to deeply detect semantic-behavioral consistency between skill documentation and execution logic. By simulating interaction trajectories of cross-skill collaboration based on risk link policies, it

identified inter-skill combinatorial risks where individually benign skills might trigger security threats during cascaded calls. In addition, Tie et al. [278] motivated stronger provenance verification and behavioral vetting for third-party skill artifacts.

Note: Skill Risks and Capability–Safety Trade-off

Skill risks demonstrate that autonomous interactive agents increasingly externalize their behavioral intelligence through reusable skill rather than relying solely on model parameters. Unlike conventional tools providing isolated functionalities, skills encapsulate executable behavioral policies, planning routines, and decision logic that become integral to long-horizon execution. From the perspective of Agentic COB, externalizing capabilities through reusing skills improves adaptability and scalability but simultaneously shifts behavioral trust from the agent itself to third-party. Ensuring behavioral safety requires not only verifying functional correctness but also validating semantic consistency, behavioral provenance, and compositional interactions among skills, so that capability reuse does not compromise long-term behavioral integrity.

6 System Safety

Community safety examines how to maintain trustworthy behaviors among interacting agents through community-level governance mechanisms. However, autonomous interactive agents do not operate in isolation from the computing infrastructure that supports their execution. Every interaction, memory update, tool invocation, and collaborative decision is ultimately mediated by the underlying runtime environment, where failures in isolation, permission boundaries, authorization, execution control, or resource management may undermine behavioral safety regardless of how well the community is governed. Consequently, ensuring reliable autonomous interactive agents requires extending the discussion beyond inter-agent governance toward system-level safety mechanisms that protect the execution environment itself, as illustrated in Figure 8.

6.1 Resource Exhaustion

Traditional denial-of-service attacks target raw infrastructure—CPU, memory, or bandwidth—but in autonomous interactive agents, computation, cost, and latency are tightly coupled to token consumption [304]. Adversaries encode multi-step protocols, detailed instructions, or hidden behavioral directives within skill descriptions or tool responses, forcing agents into extended reasoning loops, repetitive tool calls, and cascading data amplification. The result is continuous, multi-turn token drain that directly degrades the agent’s behavioral safety by diverting execution away from user goals and toward attacker-controlled resource sinks [52]. There are several relevant research efforts. ThinkTrap [158] mapped discrete tokens into a continuous agent embedding space and performed efficient black-box optimization to search for adversarial prompts, which induced black-box models to produce extremely long or infinite outputs, thereby saturating GPU memory and computational resources and achieving denial-of-service. Clawdrain [52] injected instructions into SKILL.md that induced the agent to execute a multi-turn segmented verification protocol with PROGRESS/REPAIR/TERMINAL response signals. This protocol forced the agent to generate lengthy calibration sequences across multiple rounds of tool calls. Furthermore, Dong et al. [52] revealed a costly failure phenomenon that when a skill malfunctioned, the agent autonomously entered an extended recovery loop and attempted a series of fallback options. Such autonomous error-correction behavior accumulated a massive number of tokens in the session history, causing even more severe resource depletion than when an attack succeeded. Beyond this, in multi-agent system, CORBA [402] employed malicious prompts to induce agents into a blocking state of infinite recursion, exploiting its contagious properties to propagate malicious instructions to all neighboring agents. This resulted in all reachable nodes within the multi-agent system being infected, leading to resource exhaustion and service paralysis across the system.

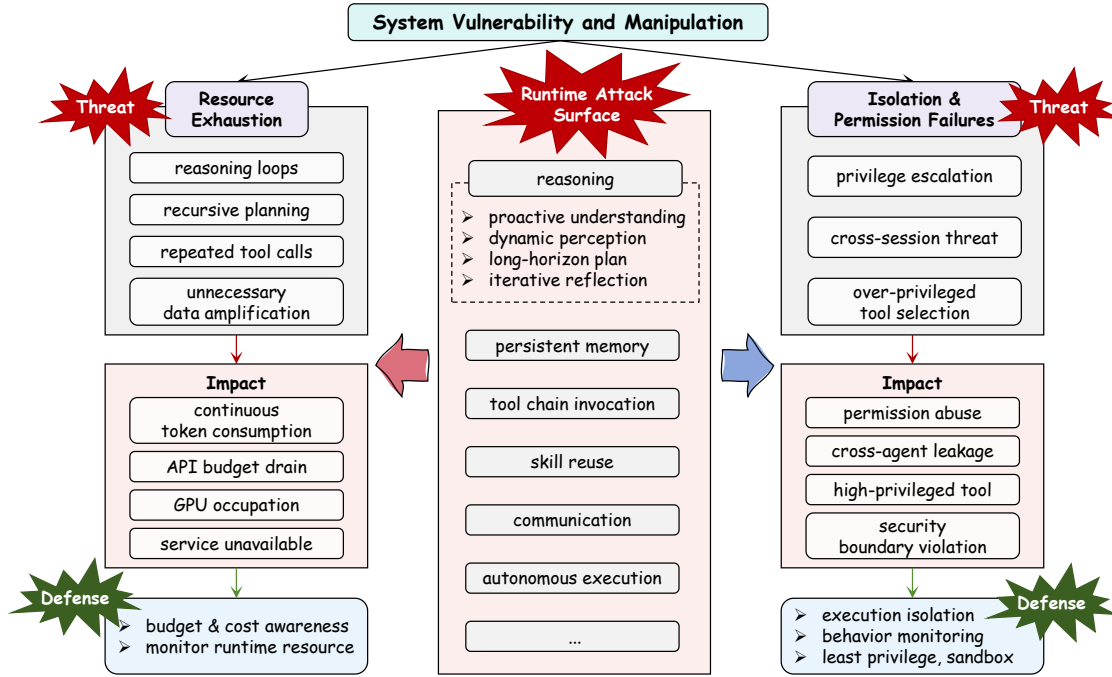


Fig. 8. The system safety of autonomous interactive agents on threat, attack, and behavioral defense.

Note: Resource Exhaustion and Capability-Safety Trade-off

Resource exhaustion attacks indicate that behavioral safety depends not only on whether an agent makes correct actions but also on whether it can sustain autonomous execution under bounded computational resources. Unlike traditional denial-of-service attacks that directly consume infrastructure resources, these attacks exploit the agent's own reasoning, planning, and recovery mechanisms to amplify token consumption, execution latency, and cascading tool interactions. This phenomenon reflects the trade-off revealed by proposed Agentic COB: increasing deliberation depth and autonomous problem-solving capability often comes at the cost of reduced behavioral sustainability under adversarial conditions. Future autonomous interactive agents should incorporate resource-aware planning, execution budgeting, adaptive scheduling, and trajectory-level resource governance during runtime, ensuring that stronger autonomy remains computationally sustainable without degrading long-horizon behavioral reliability.

Behavioral Safety Defense. Existing defenses against agent resource exhaustion risks adopt resource-aware control mechanisms, improving system resilience through budget management, execution optimization, and lifecycle-level security enforcement. To endow agents with budget awareness, Liu et al. [173] introduced a lightweight plugin, Budget Tracker, which presented perceived resource information of tool calls in real time within the reasoning loop. Building on this, their BATS framework dynamically adjusted the breadth and depth of search via a planning module to match the current budget, and employed a self-verification module to flexibly decide whether to continue along the current path or switch to a more efficient alternative, based on real-time remaining resources. For multi-agent systems, Qiu et al. [233] proposed resource-aware Co-saving to optimize

both execution efficiency and resource consumption. The Co-saving framework mined shortcuts from historical successful trajectories to bypass redundant intermediate reasoning, and utilized a filtering system that combined value, cost, and a dynamic emergency factor to select the optimal path in real time. Furthermore, it was pointed out that defending against resource exhaustion attacks required system-level control over tool composition, context management, and execution scheduling, rather than relying solely on per-call token limits or the correctness of the final output [52], which may guide future multi-architecture safety defense and prevention. AGENTWARD [385] proposed a lifecycle security architecture, extending the scope of security boundaries from content security to the realm of system security. It established system-level constraints on the agent's capability invocation, resource access, and execution behaviors via embedding layered security control mechanisms across the stages of initialization, input processing, cognition, memory, decision-making, and execution.

6.2 Isolation and Permission Failures

In autonomous interactive agents, isolation and permission boundaries are not static configuration artifacts but active safeguards that must hold across long-horizon, multi-tool, and multi-session trajectories. Failures in these boundaries—privilege escalation, over-privileged tool selection, and cross-session threat accumulation—translate directly into behavioral safety violations. An agent that exceeds its necessary permissions or leaks state across sessions does not simply malfunction but actively extends its attack surface over time, enabling attackers to progressively hijack its execution flow. This distinguishes contemporary agents from earlier tool-augmented agents, where permission was typically a per-call concern rather than a continuous behavioral property.

Specifically, adhering to the principle of least privilege, Ji et al. [114] defined privilege escalation attacks as any agent behavior that exceeded the minimum necessary to fulfill the user's intent. This manifested as the execution of unnecessary tool calls or cross-agent communication induced by external malicious input, encompassing attack scenarios such as prompt injection, RAG poisoning, untrusted agents, and the confused deputy attack in MAS. For example, *hackingBuddyGPT* [90] interacted with a target system via SSH, employing a next-cmd prompt template to guide the model in executing the next privilege escalation command, such as running system commands or testing credentials. To enable small local models to perform safety tasks under strict resource budgets, Normann et al. [214] post-trained a 4B-parameter open-weight model using Supervised Fine-Tuning and Reinforcement Learning with Verifiable Rewards, resulting in *PrivEsc-LLM*, a model specialized in executing Linux privilege escalation tasks. Furthermore, Probst et al. [226] proposed an automated Linux privilege escalation method targeting small local agents. By restructuring the attack logic from a blind "act-act" pattern into a "reflect-act" loop incorporating intervention mechanisms such as reflective analysis, Chain-of-Thought, and RAG, this approach forced the model to explicitly analyze the execution results of the previous step before making a decision, thereby overcoming common bottlenecks such as instruction repetition, hallucination, and a lack of specialized expertise. Over-privileged tool selection refers to the behavior where an agent chooses or escalates to a high-privilege tool even when a sufficiently low-privilege alternative exists. It has been found that such unnecessary expansion is common in mainstream LLM agents and is further amplified by transient failures, increasing the potential blast radius of errors, misuse, or damage [346]. Cross-session threats disperse malicious intent across multiple sessions, rendering defense systems that judge individual steps without memory incapable of identifying the overall attack trajectory [11, 95].

Behavioral Safety Defense. Recent studies mitigate agent privilege risks through mechanisms spanning least-privilege access, system isolation, runtime policy enforcement, and trusted execution control. Yang et al. [346] introduced a privilege-aware post-training defense that trained agents to preferentially use low-privilege tools when they sufficed for the task and to treat privilege escalation only as a last resort when handling failures. To further prevent autonomous interactive agents from privilege escalation attacks, subsequent research has adopted system-level isolation approaches that strictly separate the permission boundaries of different applications or

agents. IsolateGPT [325] employed a hub-and-spoke architecture to enforce execution isolation. It confined each third-party application within an independent spoke equipped with a dedicated LLM instance and storage. A trusted intermediary hub and an inter-spoke communication protocol strictly mediated and filtered all interactions between applications, ensuring the controllable safe collaboration between mutually distrusting applications. However, IsolateGPT exhibited performance degradation in multi-tool execution tasks. To address this, Ji et al. [114] proposed SEAgent, which maintained a directed graph termed the System View to capture information flows within the agent. When a new tool call was detected, a decision engine matched the real-time view against stored security policies to block the call or issue a warning. It also adopted an SEMemory module to retain context across execution rounds while mitigating risks such as context pollution. Wang et al. [290] proposed AgentSpec, the first runtime enforcement framework that intercepted key execution stages of the agent's decision pipeline. Through user-defined structured rules, the framework provided real-time, deterministic, and highly interpretable safety boundary constraints. Nevertheless, AgentSpec required expert-crafted rules for every deployment. Thus, Zhao et al. [394] proposed ClawGuard that automatically derived task-specific access rules from the user's original objective prior to any external tool invocation. By enforcing a user-confirmed rule set at every subsequent tool-call boundary, it transformed unreliable alignment-dependent defenses into a deterministic and auditable mechanism that intercepted adversarial tool calls before any real-world effect was produced. Meng et al. [197] proposed CELLMATE, which intercepted requests triggered by the agent's UI operations at the HTTP layer. It introduced an agent sitemap that mapped ambiguous low-level UI actions to HTTP requests with clear semantics, thus bridging the semantic gap and enforcing deterministic, least-privilege interception at the HTTP layer. Furthermore, Prompt Flow Integrity [131] least-privilege isolation by decoupling the system into a trusted agent that handled trusted data and a low-privilege untrusted agent. It replaced untrusted data with unique data IDs for secure content-blind reference and used real-time guardrails, which triggered mandatory user auditing whenever untrusted data attempted to invoke privileged operations or transform into instructions.

Note: Isolation Failures and Capability–Safety Trade-off

Isolation and permission failures highlight a behavioral trade-off in autonomous interactive agents: as agents evolve toward longer execution horizons, richer external interactions, and stronger autonomy, their effective permission boundary also becomes increasingly dynamic, making behavioral safety inseparable from capability evaluation. Existing defenses have moved beyond preventing individual privilege escalation events toward enforcing least-privilege execution, runtime isolation, and cross-session behavioral consistency throughout the entire interaction trajectory. From the perspective of Agentic COB, these mechanisms aim to preserve high autonomy and interaction capability while minimizing accumulated behavioral risk, thereby preventing capability improvements from proportionally enlarging the attack surface. This evolution also indicates that permission management is no longer merely an access control problem but a continuous behavioral property that must remain stable throughout long-horizon execution, providing an important foundation for building safe, trustworthy, and resilient autonomous agents.

In addition, to address the challenge that agent security systems struggle to identify cross-session threats due to a lack of stateful memory, Azarafrooz et al. [11] proposed Coreset Memory Reader. It ranked message segments via a bounded buffer situated between the message stream and a correlator, selected and retained only the highest-signal cross-session segments for submission to downstream models, enabling real-time threat detection without re-reading the complete history. He et al. [95] proposed a system-level anomaly detection framework tailored for MAS, which combined structural modeling with runtime behavioral supervision. This method modeled MAS execution as a dynamic interaction graph, where agents and tools were represented as nodes, and communications, tool calls, or reasoning steps were captured as typed directed edges while paths

through the graph then corresponded to high-risk execution trajectories or attack chains. They also developed a pluggable supervisory agent, SentinelAgent, which supported real-time execution tracing, semantic anomaly detection, and security policy intervention in a dynamically updated MAS environment, demonstrating the capability to detect distributed or covert faults and to explain anomalous behaviors.

7 Evaluation

The rapid evolution of autonomous interactive agents has significantly improved their ability to accomplish long-horizon tasks through perception, reasoning, memory, tool use, and skill recall. However, recent empirical studies consistently demonstrate that increasing task capability does not necessarily lead to safer active behaviors. Instead, stronger autonomy and richer environmental interactions expose a broader spectrum of behavioral failures that cannot be adequately characterized by conventional task success metrics alone. Consequently, recent evaluation efforts have gradually shifted from measuring model capability toward systematically assessing behavioral safety throughout the entire execution process.

As shown in Figure 9, we summarize two representative observations that motivate this transition toward intrinsic behavioral safety. Even in completely benign environments without adversarial manipulation, state-of-the-art mobile agents still exhibit a considerable benign failure rate caused by intrinsic capability limitations, including incorrect environmental understanding, unstable reasoning, improper action planning, and operation errors [28]. More importantly, although frontier foundation models continue to achieve increasingly high task utility scores, their security awareness remains substantially lower, revealing a persistent capability–security gap in modern autonomous agents [153]. Many agents successfully accomplish user objectives while simultaneously failing to recognize malicious instructions, identify suspicious interaction trajectories, or proactively reject unsafe operations. This discrepancy demonstrates that behavioral safety is not an automatic by-product of stronger reasoning capability, and utility-oriented evaluation alone cannot accurately reflect the trustworthiness of autonomous agents deployed in real environments. Figure 10 further illustrates that safety deficiencies also emerge at the trajectory level rather than only at the final task outcome [327]. Although most operating-system agents (OS agents) successfully execute a substantial proportion of individual actions, a considerable fraction of interaction trajectories still deviates toward distraction or invalid execution under unsafe scenarios especially real-world anomalies. These observations imply that evaluating only the final task completion masks many intermediate behavioral failures that accumulate throughout autonomous execution. Therefore, intrinsic behavioral safety evaluation should explicitly model the entire interaction trajectory, including perception, reasoning, planning, action execution, environmental feedback, and recovery from unexpected situations.

Further extending to extrinsic attack against autonomous interactive agents, recent empirical results demonstrate that behavioral risks are substantially amplified once agents are exposed to adversarial environments, persistent memories, third-party tools, reusable skills, and over-permissive system interfaces. It is worth noting that the model names reported in the figures denote the backbone or brain model of the corresponding autonomous interactive agent, rather than standalone chat models. Therefore, these results should be interpreted as agent-level behavioral outcomes produced by different model-driven agent systems. As shown in Figure 11, long-horizon attacks can effectively manipulate agents through memory poisoning, task injection, and backdoor mechanisms. The memory poisoning and task injection results from AgentLAB show that agents powered by representative frontier and open-weight models remain highly vulnerable to persistent-context and workflow-level manipulation [117]. Under memory poisoning, ASR ranges from 34.6% to 67.3% across tested backbones, indicating that persistent memory becomes a durable attack channel once malicious information is stored and later retrieved as trusted context. Task injection further exposes the weakness of agents under multi-turn environmental manipulation: agents powered by Qwen-3, Llama-3.1, and GPT-4o exhibit high attack success under task-injection settings, while even stronger models only reduce, rather than eliminate, the risk. In addition,

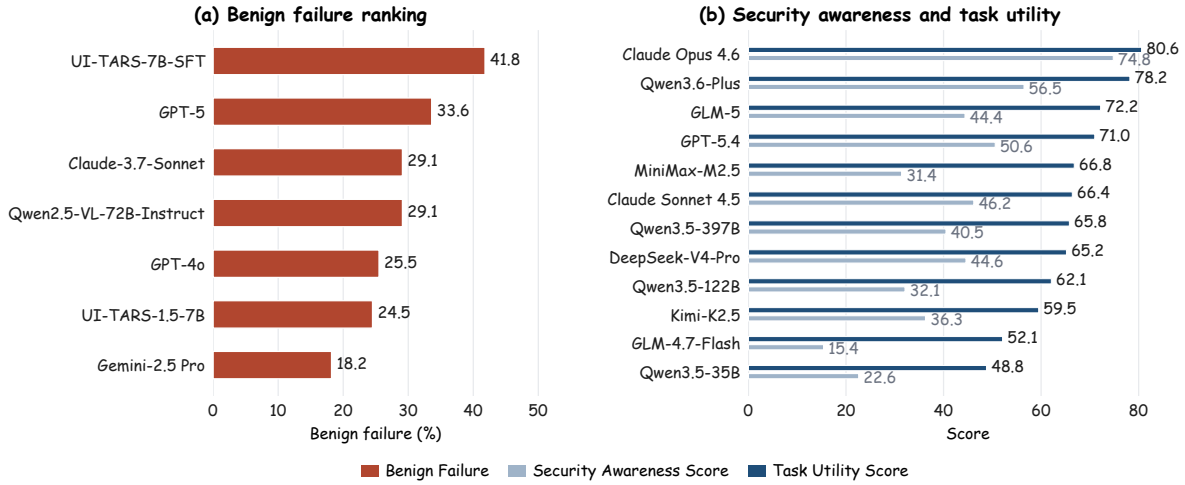


Fig. 9. Benign failure and the capability–security gap in autonomous interactive agents. (a) Benign failure refers to task failures caused by an agent’s capability limitations rather than adversarial attacks. (b) Security awareness score evaluates an agent’s ability to identify risks, anticipate unsafe trajectories, and respond appropriately, while task utility score measures task completion capability. The gap between the two indicates that stronger task performance does not necessarily imply safer behavior.

the backdoor evaluation performed by Feng et al. [65] further shows that hidden triggers can activate unsafe behaviors across different agentic attack settings. Certain agent backbones, such as GPT-5.2 and Gemini-3-Pro, exhibit high ASR across several backdoor settings, while Gemini-3-Flash is especially vulnerable on multiple attack types. These observations suggest that extrinsic attacks do not merely exploit a single unsafe response, but instead target the agent’s temporal execution process by corrupting memory, inserting malicious subtasks, or activating hidden triggers during otherwise normal workflows.

The community ecosystem introduces another important risk. Figure 12 shows that MCP poisoning produces consistently high ASR for agents powered by popular models, reaching 70.9% for DeepSeek-R1 and 61.8% for GPT-4o-mini [309]. Since MCP servers and tools are increasingly used as reusable infrastructure for agent applications, these results imply that malicious or compromised tool metadata can become a scalable supply-chain attack surface. Adversarial skill exhibits a related but more subtle failure mode. Even when the user request remains benign, unsafe influence may reside in skill guidance, helper scripts, local artifacts, runtime configurations, memory traces, or retrieval corpora [121]. Agents powered by GLM-5.1, Kimi-K2.5, Gemini-3-Flash, and GPT-5.5 still show substantial overall ASR, while maintaining non-trivial task success, which means that the agent may appear helpful and task-oriented from the user’s perspective, yet its execution path can be silently redirected by adversarial skill-facing materials. Such results reveal that community-contributed skills, MCP servers, and local execution artifacts should not be treated as inherently trustworthy extensions of the agent. System-level configuration further enlarges the behavioral safety problem. Figure 13 reports over-privileged tool use under three common forms of excessive authority: authority escalation, safety bypass, and data over-exposure. Agents powered by LLaMA-3.1-8B show particularly high over-privileged behavior, with OPUR reaching 72.7% for authority escalation, 74.1% for safety bypass, and 49.5% for data over-exposure [346]. Other backbones, including Qwen3.5-397B and DeepSeek-v3.2, also show substantial over-privileged tool selection in at least one category. These results indicate that system safety cannot rely solely on the agent’s internal restraint. When agents are

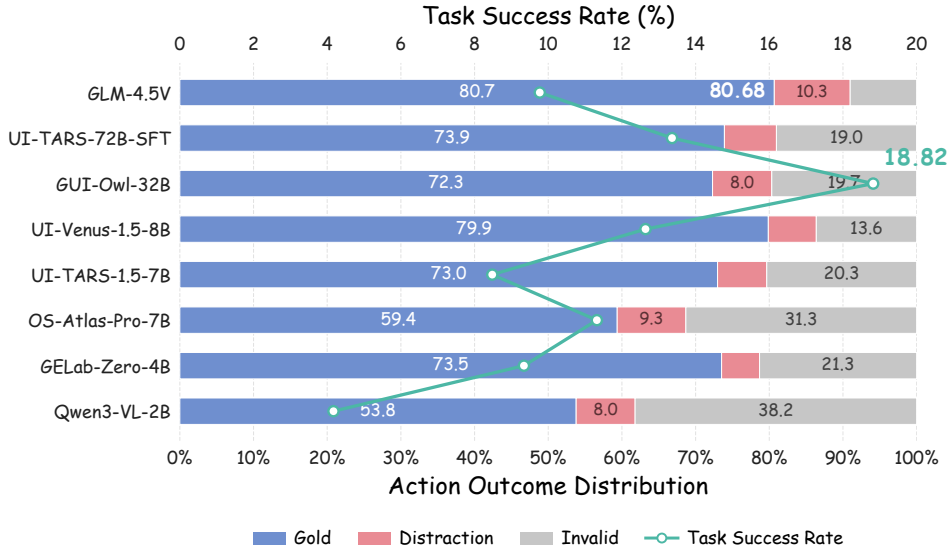


Fig. 10. Trajectory-level behavior of OS agents under unsafe scenarios. The Gold metric is incremented when the OS agent produces the correct action, Distraction is incremented when the agent follows the adversary’s intended action, and Invalid is incremented when neither case applies. Trajectory Success Rate measures successful completion across the entire interaction trajectory [327].

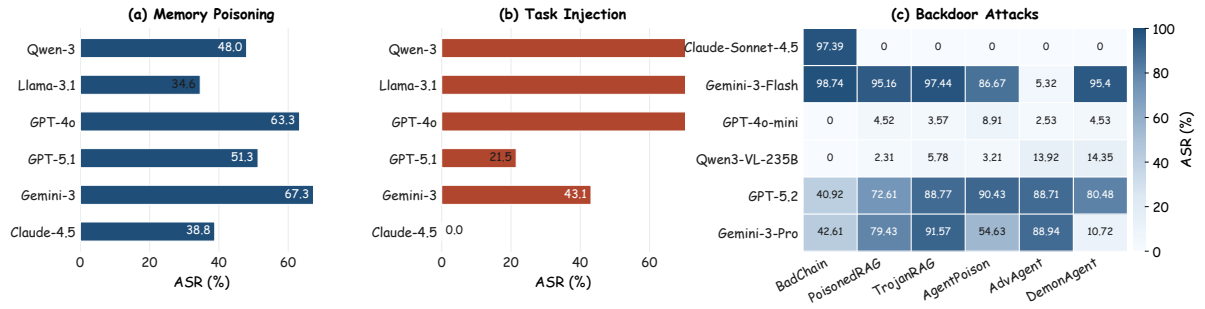


Fig. 11. Attack Success Rates of extrinsic attacks against autonomous interactive agents powered by different backbone models. (a) Memory poisoning attacks manipulate persistent memory to influence later agent decisions. (b) Task injection attacks insert malicious subtasks into otherwise normal long-horizon workflows [117]. (c) Backdoor attacks implant hidden triggers into LLM-based agents and activate unsafe behaviors under specific attack settings [65].

granted broad tool permissions, file access, or high-risk APIs, they may select unnecessarily powerful tools even when lower-privilege alternatives are sufficient. Thus, over-authorization becomes an independent source of behavioral risk, increasing the potential damage of any compromised decision.

Defense evaluation further confirms that current safeguards remain fundamentally insufficient for autonomous interactive agents. As shown in Figure 14, guard models exhibit very low detection rates on early decomposed steps in AgentHazard [63]: at round 1, no guard model exceeds 5%, which confirms that individual attack

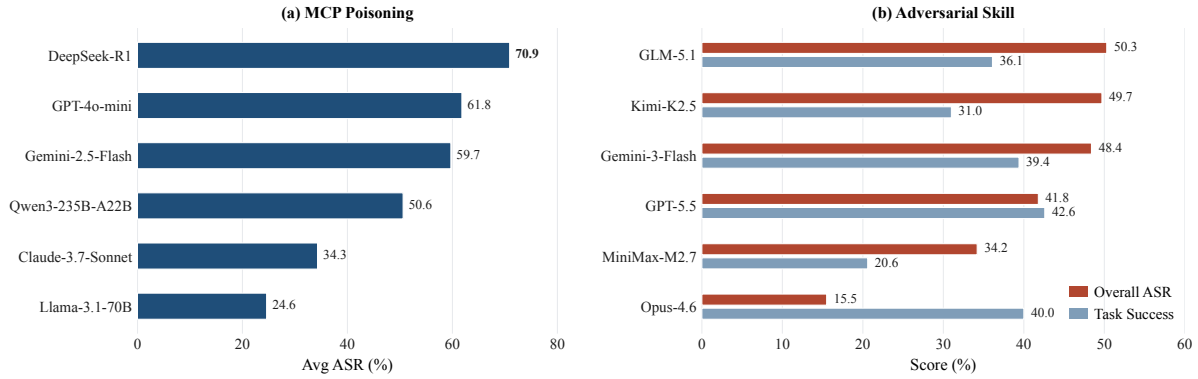


Fig. 12. Community-level attack risks in agent ecosystems. (a) Average ASR of MCP poisoning attacks against agents powered by popular backbone models, revealing widespread vulnerability to poisoned tool metadata in realistic MCP scenarios [309]. (b) Adversarial skill evaluation in realistic skill-mediated workflows, where the user request remains benign but skill-facing guidance, local artifacts, or execution-environment files can steer the agent toward unsafe behavior. ASR denotes the fraction of evaluated cases in which the intended unsafe behavior is observed [121].

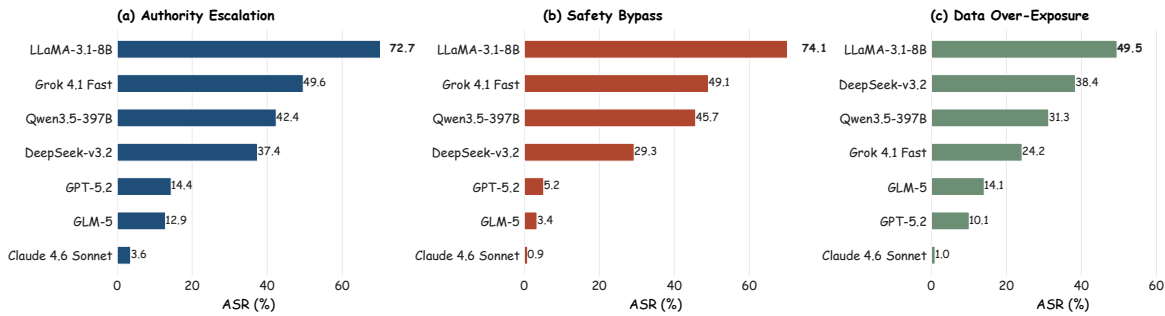


Fig. 13. System-level behavioral safety risks caused by over-privileged tool use. The Over-Privileged Tool Use Rate (OPUR) of agents powered by different backbone models across three frequent forms of over-privileged behavior: authority escalation, safety bypass, and data over-exposure [346].

steps often appear locally benign. Detection improves as more steps are concatenated, but remains consistently limited even at round_all. The best model, Llama-Guard-3-8B, reaches only 27.03%, while the Qwen3Guard family plateaus around 15–17%. The near-identical performance of Qwen3Guard-0.6B, Qwen3Guard-4B, and Qwen3Guard-8B suggests that the main bottleneck is not merely model capacity, but a mismatch between existing guard training distributions and multi-step agentic harmful intent. The guarded agent results also reveal a persistent safety–utility tension: detect-and-abort style defenses can reduce ASR, but often degrade benign task completion, whereas feedback-based methods improve the trade-off but still do not provide a general solution across benchmarks and attack types [202].

Taken together, the evidence forms a coherent progression from intrinsic behavioral risk to extrinsic and systemic risk. Intrinsic safety evaluations show that agents can fail even in benign environments because high task utility does not imply strong security awareness. Extrinsic attack evaluations show that adversaries can exploit the agent’s memory, retrieval, long-horizon planning, and environmental feedback to induce unsafe

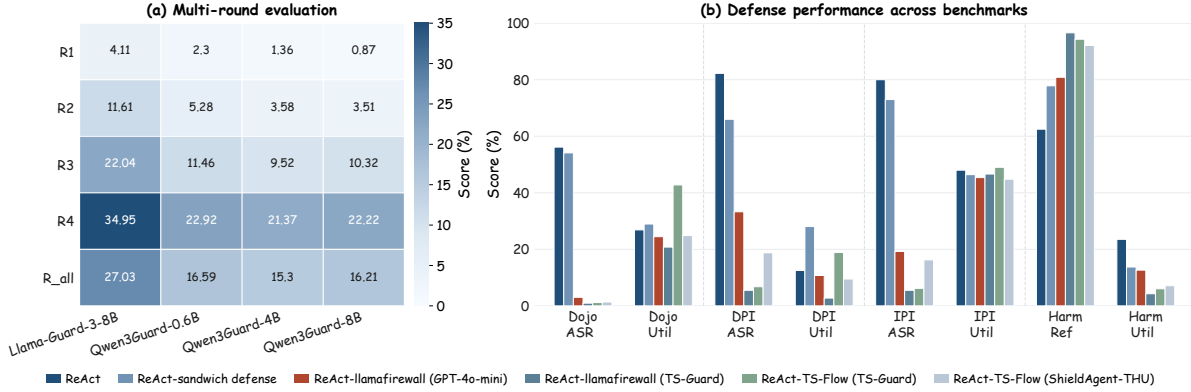


Fig. 14. Defense performance against agentic attacks. (a) Unsafe Detection Rates of guard models on AgentHazard. Detection is very limited for individual decomposed steps and improves only partially as more steps are concatenated, remaining insufficient even at round_all [63]. (b) Evaluation of guarded agents across multiple benchmarks, illustrating the trade-off between safety and utility under direct prompt injection, indirect prompt injection, and harmful-task settings [202].

behavior. Community-level evaluations demonstrate that poisoned MCP servers, tools, and skills can compromise agents through components that appear useful and legitimate. System-level evaluations further show that over-privileged tool access magnifies the consequence of unsafe decisions. Meanwhile, existing defenses remain limited, especially against multi-step, locally benign, and context-dependent attacks. These findings collectively indicate that behavioral safety is a central requirement for autonomous interactive agents: trustworthy deployment requires evaluating and defending the entire perception-reasoning-planning-action trajectory, rather than only filtering isolated prompts or final responses.

Following the behavioral risk taxonomy presented in this survey, we organize existing evaluation benchmarks into two complementary categories. Intrinsic safety benchmarks evaluate the robustness and resilience of autonomous interactive agents against endogenous behavioral failures and potentially unsafe execution environments, measuring whether the agent can maintain secure decision-making throughout the complete behavior operation. In contrast, extrinsic attack benchmarks evaluate the effectiveness of adversarial attack strategies, focusing on their effectiveness to manipulate agent perception, reasoning, memory, planning, or action execution to induce unsafe behaviors. They characterize both sides of the behavioral safety landscape, namely the defensive reliability of autonomous interactive agents and the offensive capability of emerging agent-specific attacks.

7.1 Intrinsic Safety Benchmarks

Intrinsic benchmarks focus on failures that arise from the agent’s own decision process, including unsafe compliance, poor risk awareness, hallucinated execution, and inability to recover from hazardous intermediate states. For computer-use agents, OS-HARM [136] evaluates safety in realistic operating-system environments by asking agents to handle harmful tasks, prompt-injection-like situations, and risky benign workflows, measuring how often agents execute unsafe actions rather than merely whether they refuse at the text level. AgentHazard [63] similarly targets computer-use agents, but emphasizes execution-level failures that accumulate across multi-turn trajectories, including security-relevant behaviors such as data exfiltration, destructive actions, privilege escalation, and unsafe code execution. LPS-Bench [34] uncovers critical long-horizon, planning-time safety vulnerabilities of MCP-based computer-use agents (CUAs) across diverse computer-use scenarios, characterizing where and how safety risks arise during long-horizon planning. TS-Bench [202] complements these evaluations

Table 2. Intrinsic Safety Benchmarks.

Benchmark	Domain	Behavior Aware	Multi Turn	Reliability Metric	Keypoints
OS-HARM [136]	computer-use agents	✓	✓	Unsafe Rate	harmful tasks
AgentHazard [63]	computer-use agents	✓	✓	ASR, Score	execution-level failures
IS-Bench [182]	embodied household	✓	✓	SR, SSR	risk mitigation
ATBench [157]	realistic trajectory	✓	✓	Acc, F1, Recall	diverse risk patterns
ATBench-Claw & ATBench-Codex [355]	OpenClaw & Codex runtime settings	✓	✓	Acc, F1, Recall	customized taxonomy
Claw-Eval [358]	real-world	✓	✓	Score, Pass@k	full-trajectory auditing
Risky-Bench [397]	daily life-assist	✓	✓	ASR	domain-agnostic safety
LPS-Bench [34]	computer-use agents	✓	✓	SR	planning-time safety awareness
MIRAGE-Bench [384]	interactive scenarios	✓	✓	US, HR	hallucinations
EU-Agent-Bench [163]	EU law	✓	✗	Legality Rate	alignment with legal norms
AGENTS SAFE [365]	embodied agents	✓	✓	GR, HR, TSR	multi-stage evaluation
OPENAGENTS SAFETY [282]	real-world	✓	✓	Unsafe Behavior Rate	modular & extensible
MobileSafetyBench [141]	realistic mobile devices	✓	✓	Goal Achievement Rate	helpfulness and safety
SafeMindBench [33]	embodied agents	✓	✓	SR, SSR, ER	stage isolation and realism
AgentCanary [153]	real execution	✓	✓	OSS, SAS, TUS	trajectory-grounded
Replibench [21]	capability-based tasks	✓	✓	Pass@k, SR	autonomous replication
AgentHarm [9]	malicious tasks	✓	✗	HS, RR	harmfulness and jailbreaks
ASSEBench [187]	real safety & security	✓	✓	Acc, F1, Recall	post-hoc trajectory-level
TS-Bench [202]	tool invocation	✓	✓	Acc, ASR, Utility	step-level detection
HINTBench [293]	realistic trajectory	✓	✓	Acc, F1, Recall	intrinsic failures

Abbreviations: Attack Success Rate (ASR), Success Rate (SR), Safety Success Rate (SSR), Accuracy (Acc), Utility Score (US), Hallucination Rate (HR), Grounding Recall (GR), Task Success Rate (TSR), Effectiveness Rate (ER), Outcome Safety Score (OSS), Security Awareness Score (SAS), Task Utility Score (TUS), and Harm Score (HS).

at the tool-invocation level, measuring whether an agent can detect unsafe tool calls while preserving task utility. Another work studies embodied and interactive agents, where safety depends on physical or simulated consequences. IS-Bench [182] evaluates VLM-driven household agents in dynamic scenarios where hazards can appear or worsen as the agent acts, using task success and safety success to capture both completion and risk mitigation. AGENTS SAFE [365] provides a multi-stage embodied-agent evaluation spanning perception, planning, and execution, showing that recognizing a hazard is not sufficient if the agent cannot translate that recognition into safe action. SafeMindBench [33] further emphasizes stage isolation and realistic embodied execution, measuring success, safety success, and effectiveness to identify where safety failures enter the pipeline. Risky-Bench [397] broadens the setting to daily life-assistance tasks and evaluates whether agents avoid unsafe actions in domain-agnostic situations where risks are not tied to a single platform or tool interface.

Several benchmarks evaluate safety from full trajectories rather than isolated prompts or final answers. AT-Bench [157] introduces realistic agent trajectories with diverse risk patterns and evaluates detection using accuracy, F1, and recall. ATBench-Claw and ATBench-Codex [355] adapt this trajectory-level idea to OpenClaw and Codex runtime settings with a customized taxonomy, enabling safety analysis under concrete agent harnesses rather than abstract traces. Claw-Eval [358] performs full-trajectory auditing in real-world tasks and

records each run through execution traces, audit logs, and environment snapshots to enable trajectory-aware grading. ASSEBench [187] similarly treats agent safety and security as post-hoc trajectory-level auditing problems, evaluating whether models can recognize safety-critical events after observing realistic execution traces. HINTBench [293] focuses on intrinsic failures in realistic trajectories, using classification metrics to evaluate whether evaluators can detect harmful or unreliable agent behavior across multi-turn traces. AgentCanary [153] is also trajectory-grounded, measuring outcome safety, security awareness, and task utility in real execution settings where the evaluator must judge not only whether the agent succeeded, but whether it understood and respected safety constraints. Other intrinsic benchmarks cover specialized dimensions of agent reliability. MIRAGE-Bench [384] evaluates hallucinations in interactive scenarios, using utility and hallucination-oriented metrics to expose cases where agents fabricate observations, actions, or environmental states during execution. EU-Agent-Bench [163] studies alignment with legal norms under EU law, measuring legality rate and highlighting that safe behavior may require compliance with jurisdiction-specific constraints rather than generic refusal rules. OPENAGENTSAFETY [282] provides a modular and extensible benchmark for real-world agent safety, reporting unsafe behavior rates across multi-turn tasks. MobileSafetyBench [141] transfers evaluation to the helpfulness and safety of agents controlling mobile devices in realistic interactive environments, observing that agents exhibit unsafe behaviors in many daily scenarios, which highlights the urgent need for continued research to develop more robust safety mechanisms in mobile environments. Replibench [21] evaluates whether agents can autonomously replicate and persist in the wild, covering four core domains: obtaining resources, exfiltrating model weights, replicating onto compute, and persisting on this compute for long periods. Finally, AgentHarm [9] measures how likely and how capable agents are of completing harmful tasks when they must execute multi-step tool calls to fulfill user requests.

7.2 Extrinsic Attack Benchmarks

Extrinsic attack benchmarks have evolved from evaluating isolated prompt injection attacks toward comprehensive assessment of adversarial agent ecosystems. Recent benchmarks increasingly consider long-horizon interactions, dynamic environments, tool ecosystems, MCP infrastructures, skill supply chains, multimodal interfaces, and multi-agent protocols, providing increasingly realistic evaluations of security risks faced by autonomous interactive agents operating in open-world environments. RSB [373] systematically evaluates poisoning attacks against diverse RAG architectures by measuring the influence of malicious retrieved documents on downstream agent behaviors, while PoisonArena [30] introduces competing poisoning attacks, where multiple adversaries with conflicting objectives simultaneously manipulate the same query, enabling more realistic robustness evaluation. These benchmarks are important because many agents rely on retrieved context as operational memory, making the retrieval channel an indirect control surface.

Prompt injection remains the dominant attack class for tool-using and web agents. WInjectBench [179] evaluates prompt-injection detection for web agents through true- and false-positive rates, focusing on whether malicious web content can be distinguished from benign page content before it influences behavior. WASP, AGENTPI, AgentDyn, and BrowseSafe-Bench evaluate prompt injection in increasingly realistic agent settings. BrowseSafe-Bench [381] embeds attacks in richly structured HTML payloads, reflecting the complexity of real-world browser agent interactions; WASP [60] simulate in different web environments in a realistic manner within a sandbox web environments; AGENTPI [298] emphasizes context-dependent tasks where malicious instructions become effective only in relation to the user goal; and AgentDyn [148] further requires dynamic planning and incorporates helpful third-party instructions throughout the execution trajectory, enabling a more adaptive and realistic evaluation. INJECAGENT [370] studies indirect prompt injection in tool-integrated agents, where untrusted tool outputs induce malicious actions while the user request remains benign. AgentDojo [47] provides a dynamic tool-calling environment with realistic user tasks, injection tasks, attacks, and defenses,

Table 3. Extrinsic Attack Benchmarks.

Benchmark	Domain	Behavior Aware	Multi Turn	Reliability Metric	Keypoints
RSB [373]	RAG architectures	✓	✗	Acc, ASR, F1	RAG poisoning
PoisonArena [30]	RAG	✓	✗	ASR, F1	competing poisoning attack
WAIInjectBench [179]	web agents	✗	✗	TPR, FPR	prompt injection detection
WASP [60]	web agent	✓	✓	ASR, Utility	prompt injection
AGENTPI [298]	context-dependent tasks	✓	✓	ASR, Utility	prompt injection
AgentDyn [148]	dynamic agents	✓	✗	ASR, Utility	prompt injection
BrowseSafe-Bench [381]	browser agents	✓	✗	Acc, F1, Recall	prompt injection
SKILL-INJECT [246]	skill files	✓	✗	ASR, Utility	prompt injection
VPI-Bench [23]	web agents	✓	✗	SR, AR	visual prompt injection
INJECAGENT [370]	tool-integrated agent	✓	✗	ASR, Utility	indirect prompt injection
ASB [378]	agent operation	✓	✗	ASR, RR, NRP	various attacks & defense
AgentDojo [47]	tool-calling	✓	✗	ASR, Utility	prompt injection robustness
MSB [376]	full tool-use pipeline	✓	✓	ASR, PUA, NRP	MCP-specific attacks
PASB [306]	personalized AI	✓	✓	ASR, Resp Rate	black-box end-to-end
A2ASecBench [156]	A2A-MAS	✓	✓	ASR, Utility	supply-chain, protocol-logic
GhostEI-Bench [28]	mobile ecosystems	✓	✓	TC, BF, VR	dynamic environment injection
AgentLAB [117]	realistic scenarios	✓	✓	ASR, T2S, Utility	adaptive long-horizon attacks
SkillSafetyBench [121]	skill workflows	✓	✓	ASR, TSR	skill-facing attacks
MCP-SafetyBench [409]	real MCP servers	✓	✓	ASR, TSR	multi-turn robustness
HarmfulSkillBench [120]	realistic ecosystems	✓	✓	Refusal, Score	harmful skills
ClawTrojan [266]	local agentic harnesses	✓	✓	ASR, Penetration	multi-step trojan attack
A3S-Bench [191]	OpenClaw	✓	✓	TCR, RTR@k	multi-dimension evasions
MemJack-Bench [29]	multimodal	✓	✓	ASR	jailbreak attack
MCP-TDP [172]	real-world	✓	✓	ASR, UCC	tool description poisoning
MCPTox [309]	realistic MCP servers	✓	✗	ASR, RR	tool poisoning attacks
SIREN [165]	realistic environment	✓	✓	ASR, Utility	tool stream attacks
MalSkillBench [83]	malicious skill	✓	✓	Acc, F1, Recall	runtime-verified
macOSWorld [347]	multilingual interaction	✓	✓	Gold/Distracted Rate	macOS deception attacks
ART Benchmark [410]	realistic deployment	✓	✓	Behavior/Transfer ASR	red-teaming competition
AgentRedBench [51]	enterprise integrations	✓	✓	ASR, Utility	indirect prompt injection
DECOMPBENCH [133]	real-world	✓	✓	ASR, RR, CPR	decomposition attacks
TOOLPRIVBENCH [346]	realistic tool-use	✓	✓	OPUR, PED	over-privileged tool selection
Liu et al. [168]	realistic scenarios	✓	✓	SR, MR, Detection Rate	dynamic adversarial attacks
Yao et al. [356]	OpenClaw	✓	✓	AGS, UGS	contextual vulnerabilities

Abbreviations: True Positive Rate (TPR), False Positive Rate (FPR), Attempted Rate (AR), Rejection/Refusal Rate (RR), Net Resilient Performance (NRP), Performance Under Attack (PUA), Task Completion (TC), Benign Failure (BF), Vulnerability Rate (VR), Turn to Success (T2S), Task Completion Rate (TCR), Risk Trigger Rate (RTR), User Command Completion (UCC), Checkpoint Pass Rate (CPR), Over-Privileged Tool Use Rate (OPUR), Pre-Escalation Exploration Depth (PED), Misleading Rate (MR), Attack Grading Score (AGS), and Utility Grading Score (UGS).

making it a central benchmark for jointly measuring utility and prompt-injection robustness. ASB [378] broadens the attack surface to multiple agent-operation attacks and defenses, reporting attack success, refusal, and net resilient performance. Recent benchmarks extend injection attacks beyond text. VPI-Bench [23] evaluates visual prompt injection against web agents, testing whether adversarial instructions embedded in visual content can redirect agent behavior. GhostEI-Bench [28] studies dynamic environment injection in mobile ecosystems, where attackers control realistic third-party content such as messages, titles, or product descriptions rather than the system prompt or app code. macOSWorld [347] evaluates deception attacks in multilingual macOS interaction, measuring whether agents follow the intended gold path or are distracted by adversarial UI content. Liu et al. [168] study dynamic adversarial attacks in realistic mobile scenarios, using success, misleading, and detection rates to quantify whether mobile agents can resist deceptive environmental cues.

As agent ecosystems become increasingly complex, recent benchmarks target skills, tools, MCP servers, and agent supply chains. SKILL-INJECT [246] extends indirect prompt injection to reusable skill files. SkillSafetyBench [121] evaluates unsafe influence that may reside in skill guidance, local artifacts, or execution environment files that steer the agent toward unsafe actions. HarmfulSkillBench [120] studies harmful third-party skills within realistic agent ecosystems, and MalSkillBench [83] focuses on runtime verification of malicious skills using execution-aware evaluation protocols. TOOLPRIVBENCH [346] evaluate whether agents choose higher-privilege tools despite sufficient lower-privilege alternatives, measuring both aggressive selection and premature escalation. MCP-oriented benchmarks further formalize tool-protocol risks. MSB [376] evaluates attacks against the complete MCP-based tool-use pipeline, measuring both attack success and performance degradation. MCP-SafetyBench [409] builds on real MCP servers that supports realistic multi-turn evaluation, and reveals the vulnerability to MCP attacks with a notable safety-utility trade-off. MCP-TDP [172] specifically investigates tool description poisoning attacks by manipulating natural-language tool manuals, and MCPTox [309] extends this direction by evaluating poisoning attacks across realistic MCP ecosystems. SIREN [165] further introduces tool-stream attacks that exploit sequential tool interactions during execution. Several benchmarks study broader adversarial agent ecosystems. PASB [306] evaluates black-box attacks against personalized AI assistants under realistic end-to-end interaction settings. A2ASecBench [156] targets agent-to-agent communication by covering supply-chain attacks and protocol-logic vulnerabilities within multi-agent systems. AgentLAB [117] evaluates adaptive long-horizon attacks under realistic deployment scenarios such as intent hijacking, tool chaining, objective drifting, task injection, and memory poisoning. ClawTrojan [266] studies multi-step trojan attacks in local agentic execution frameworks, while A3S-Bench [191] evaluates temporal, spatial, and semantic evasion strategies against OpenClaw agents. MemJack-Bench [29] investigates multimodal jailbreak attacks exploiting memory mechanisms during long-horizon interactions. ART Benchmark [410] establishes a competitive red-teaming benchmark that measures both attack transferability and behavioral robustness under realistic deployment settings. AgentRedBench [51] evaluates indirect prompt injection attacks against enterprise-integrated agents, while DECOMPBENCH [133] investigates decomposition attacks that exploit multi-step reasoning processes to bypass safety constraints. Yao et al. [356] further analyze contextual vulnerabilities of OpenClaw agents through fine-grained grading of attack effectiveness and utility preservation.

Taken together, these benchmarks show that behavioral safety of autonomous interactive agents must be evaluated along both intrinsic safety and extrinsic attack axes. Intrinsic safety benchmarks measure robustness and resilience: whether agents can recognize risks, avoid unsafe execution, recover from hazardous intermediate states, and preserve utility under benign but safety-critical conditions. Extrinsic attack benchmarks measure adversarial effectiveness: whether attackers can exploit retrieval, web content, visual inputs, skills, tools, MCP servers, memory, or multi-agent protocols to redirect behavior. A comprehensive evaluation protocol for agent safety should report not only task success or refusal rate, but also trajectory-level safety, gold action rate, utility under attack, recovery behavior, and the stage at which failure occurs.

8 Future Work

The rapid evolution of autonomous interactive agents is fundamentally reshaping the research landscape of AI safety. Unlike conventional task-oriented tool-using agents, emerging agentic systems increasingly operate as persistent runtime services equipped with long-term memory, reusable skills, continuous environmental interaction, and autonomous capability evolution. Ensuring behavioral safety is becoming a long-horizon systems challenge rather than a static model-level problem. Looking ahead, we believe the following research directions will be essential for building trustworthy autonomous interactive agents.

Unified Behavioral Safety Modeling. Most existing safety mechanisms are predominantly component-oriented, protecting individual modules such as prompts, memories, tools, skills, or reasoning processes independently. However, unsafe behaviors rarely originate from a single component, but instead emerge from cumulative interactions among foundation models, memory systems, planning modules, communication protocols, external tools, runtime environments, and collaborating agents throughout long execution trajectories. Future research should move beyond modular protection and establish unified behavioral safety frameworks capable of modeling how risks propagate, accumulate, and amplify across the entire behavioral lifecycle. Such frameworks should jointly characterize cognitive reliability, execution robustness, interaction resilience, and behavioral safety instead of evaluating isolated safety properties.

Continuous Runtime Safety Assurance. Offline alignment and pre-execution filtering alone are insufficient for autonomous interactive agents whose internal states continuously evolve during execution. As agents dynamically update memories, revise plans, invoke external tools, and actively respond to environmental feedback, safety assurance must become an online capability throughout the execution process. This requires runtime mechanisms capable of monitoring execution trajectories, verifying behavioral intentions, estimating safety risks, detecting intention drift, and adaptively regulating permissions, memory updates, and planning decisions before unsafe behaviors escalate. Such continuous governance will become increasingly important as agents evolve into always-on digital workers executing long-duration tasks with limited human supervision.

Safe Continual Agent Evolution. Long-term memory, self-reflection, autonomous experience acquisition, and reinforcement learning are enabling agents to continuously improve after deployment. While continual evolution significantly enhances adaptability and autonomy, it also introduces new risks including memory corruption, policy drift, reward hacking, and cumulative behavioral deviations. Future agent architectures need safety-preserving continual learning mechanisms that validate newly acquired knowledge, audit policy updates, and prevent unsafe experiences from becoming persistent behavioral patterns. Balancing continual capability improvement with long-horizon behavioral stability will become a defining challenge for next-generation autonomous interactive agents.

Secure Open Agent Ecosystems. Autonomous interactive agents are increasingly deployed within open ecosystems consisting of communication protocols, MCP servers, reusable skills, external repositories, cloud runtimes, and large-scale multi-agent collaboration. Consequently, safety can no longer be guaranteed solely at the individual agent level. This calls for ecosystem-level trust mechanisms supporting behavioral provenance, capability certification, trust negotiation, secure delegation, inter-agent accountability, as well as cross-platform policy enforcement. Future safety frameworks should ensure that trustworthy behaviors remain preserved while agents dynamically discover, compose, and collaborate with previously unknown external services and heterogeneous agent communities.

Behavior-Centric Safety Evaluation. Existing benchmarks primarily assess task completion or attack success under relatively static settings, providing limited insight into behavioral reliability during long-horizon autonomous execution. Future evaluation protocols should move beyond one-shot assessment toward continual and adaptive evaluation, measuring behavioral integrity across long execution trajectories, cumulative risk evolution, recovery from unsafe behaviors, long-term memory reliability, runtime robustness, and capability-risk

trade-offs. In particular, unified evaluation methodologies inspired by the Agentic COB perspective are expected to provide a more comprehensive understanding of behavioral safety in increasingly autonomous agent systems.

Autonomous interactive agents are gradually evolving from task-oriented assistants into persistent AgentOS capable of continuously interacting with open digital ecosystems. This paradigm shift fundamentally transform AI safety from protecting individual model outputs toward governing the complete lifecycle of autonomous behaviors. We anticipate that unified behavioral safety will become one of the fundamental principles underlying the design, deployment, and evaluation of next-generation autonomous interactive agents.

9 Conclusion

The emergence of autonomous interactive agents marks a fundamental transition from traditional LLM-based assistants toward persistent intelligent systems capable of long-horizon reasoning, continual interaction, autonomous decision-making, and behavioral evolution. Compared with conventional task-oriented tool-augmented agents, their expanding autonomy, interaction capability, and ecosystem connectivity substantially enlarge the behavioral attack surface, making behavioral safety a primary challenge for trustworthy deployment.

In this survey, we presented a unified behavior-centric safety framework for autonomous interactive agents. Starting from intrinsic cognitive vulnerabilities, we systematically review behavioral risks originating from foundation models, memory mechanisms, tool invocation, and multi-agent collaboration. Building upon these inherent vulnerabilities, we further summarize extrinsic loss-of-control risks arising from supply-chain manipulation, user input, environmental interaction, and collaborating agents. Considering the increasing reliance on open infrastructures, we additionally discuss community safety and system safety involving communication protocols, reusable skills, runtime environments, permission management, and execution governance. Furthermore, we summarize representative behavioral safety defenses and organized existing evaluation benchmarks according to intrinsic behavioral safety and extrinsic behavioral attacks. Throughout the survey, we introduce Agentic COB as a conceptual perspective for understanding the capability-behavioral risk trade-off accompanying increasingly autonomous interactive agents. As future agents progressively evolve toward persistent operating AgentOS and ecosystem-level intelligent infrastructures, ensuring trustworthy proactive behaviors will require moving toward continuous behavioral governance across the entire agent lifecycle. We hope this survey provides a systematic foundation for future research on behavior safety and contributes to the development of reliable, controllable, and trustworthy autonomous interactive agents.

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